Before the Public Service Commission

The Brooklyn Union Gas Company d/b/a National Grid NY

and

KeySpan Gas East Corporation d/b/a National Grid

Rebuttal Testimony

of

Ann E. Bulkley

Case 19-G-0309
Case 19-G-0310

September 18, 2019

Rebuttal Testimony of Ann E. Bulkley

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I. INTRODUCTION

Q. Please state your name, affiliation, and business address.

A. My name is Ann E. Bulkley. I am a Senior Vice President of Concentric Energy Advisors, Inc. ("Concentric"), located at 293 Boston Post Road West, Suite 500, Marlborough, Massachusetts 01752.

Q. Are you the same Ann E. Bulkley who provided Direct Testimony in this proceeding?

A. Yes. The terms defined in my Direct Testimony have the same definitions here unless otherwise indicated.

II. PURPOSE AND OVERVIEW OF REBUTTAL TESTIMONY

Q. What is the purpose of your rebuttal testimony?

A. The purpose of my Rebuttal Testimony in these proceedings is to respond to the prepared testimony of the Department of Public Service Staff ("Staff") Finance Panel regarding the appropriate return on equity ("ROE") for KEDNY and KEDLI. My analysis is supported by the data presented in Exhibits__ (AEB-1R) through (AEB-4R).

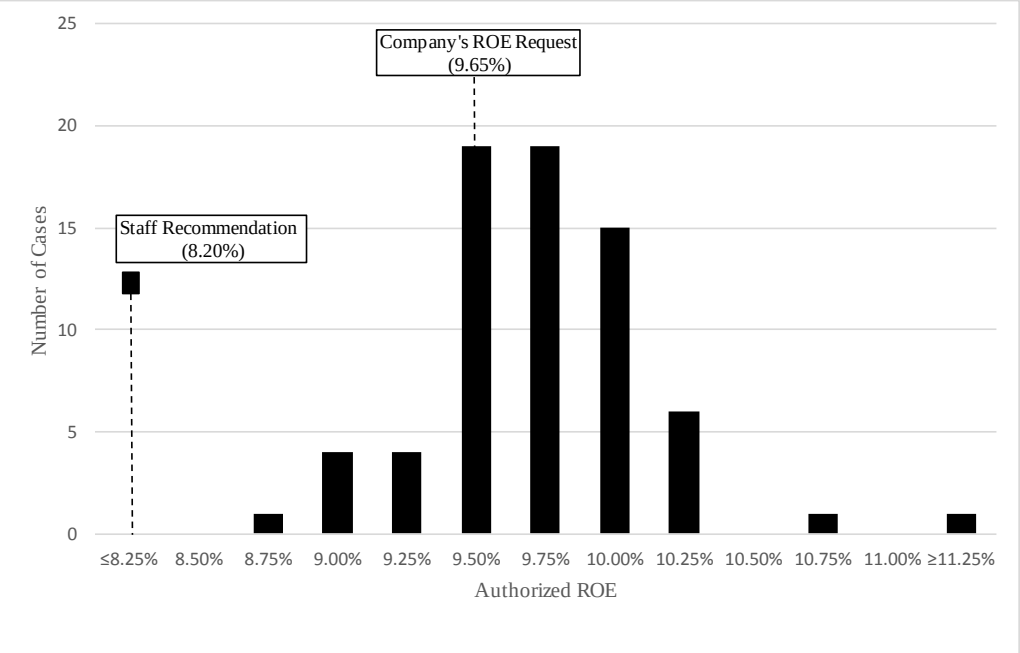
Q. Please summarize your principal concerns with the Staff Finance Panel's recommendations.

A. The Staff Finance Panel's recommended ROE of 8.20 percent is 80 basis points lower than the current authorized ROE for KEDNY and KEDLI of

9.00 percent (December 2016). Staff's recommendation is also well below any authorized ROE for a gas distribution company in the U.S. since at least 1980.[1] Staff has provided no evidence to demonstrate that the Companies' risk profiles are sufficiently different from other natural gas distribution companies to justify a return that is more than 144 basis points lower than the average ROE award for other gas distribution companies over the period from 2017 through 2019. While Staff cites the current low interest rate environment in support of its recommendation, this environment has existed for some time and affects all electric and gas utilities. As shown in Figure 1, over the last three years, the majority of the authorized ROEs have been greater than 9.65 percent.

[1] Source: S&P Global Market Intelligence (Formerly known as SNL Financial).

Figure 1: Authorized ROE for Natural Gas Distribution Utilities
January 2017 through September 2019



Staff applied a two-thirds weighting to the results of the Discounted Cash Flow ("DCF") analysis and a one-third weight to the Capital Asset Pricing Model ("CAPM") analysis, based on the specifications of the 1991 Recommended Decision ("RD") in the Generic Finance Proceeding ("GFP") and other Commission discussions of the ROE estimation methodology. Staff's entire justification for its methodology is that the Commission has used this approach in prior cases. While Staff cites to other cases, it fails to recognize that the Commission has made adjustments to its methodology since the RD, to rely on better and more current information. The Commission's methodology has evolved over time, suggesting that the Commission is open to considering whether the data and methodologies that it has historically relied upon are the best and most relevant given prevailing

market conditions and financial analysis best practices at the time of the case it is deciding.

In contrast to the Commission's demonstrated willingness to change its approach to reflect the circumstances at the time of its decision, Staff has not questioned why the result of its ROE analyses is inconsistent with authorized returns available in New York and other jurisdictions, and whether that difference is, or can be, accounted for by evidence. Rather, Staff has applied the models in a purely mechanistic fashion based on Commission "precedent" without using informed judgment or considering current economic and market conditions, as is appropriate and as the Commission has demonstrated a willingness to do in prior cases.

Furthermore, while Staff acknowledges that credit rating agencies "view unpredictability as an additional risk,"[2] they have not considered how credit rating agencies or equity investors would react to an extraordinarily low authorized ROE for the Companies relative to recently approved ROEs. Such a low ROE is likely to be viewed as punitive by rating agencies and investors and cause rating agencies to question the level of regulatory support being provided by the Commission, with consequences that could be borne by all New York utilities and their customers. As Staff points outs, there is an incremental risk due to the effect of the Companies' moratorium

[2] Prepared Testimony of the Staff Finance Panel, at 92-93.

on processing new natural gas service applications, implying that the Companies may forego growth opportunities in the future.[3] It is not clear how Staff determined that an ROE that is 48 basis points lower than the lowest authorized ROE since at least 1980 would be viewed by the investment community as predictable. More likely, such a result would be viewed as a decrease in the Commission's overall supportiveness, which Moody's identifies as a specific factor that could lead to a downgrade for the Companies.[4] In addition, it would negatively affect the Companies' "Sufficiency of Rates of Returns," subfactor which is currently rated Baa for the Companies.[5] Moreover, predictability is not the only goal of rating agencies and investors, nor is it a requirement in setting the ROE as established by the *Hope* and *Bluefield* standards.

The Staff's recommendation fails to address the concerns of credit rating agencies following tax reform. As discussed in my direct testimony, Moody's recently downgraded the senior unsecured rating and issuer rating of KEDNY and KEDLI, noting the persistent weakness of key credit metrics with limited likelihood of a recovery in the medium term. As discussed in more detail in Section VII, Moody's rationale includes the expectation of reduced revenues for the Companies due to the TCJA.[6]

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3 Prepared Testimony of the Staff Finance Panel, at 75.

4 *See* Exhibit ___ (SFP-5) at 2 and Exhibit ___ (SFP-6) at 2.

5 *See* Exhibit ___ (SFP-5) at 8 and Exhibit ___ (SFP-6) at 7.

6 Moody's Investors Service, Rating Action: Moody's downgrades KEDNY to A3; stable outlook, February 22, 2019 and Moody's Investor Services, Ratings Action:

Staff's recommendation to reduce the ROE by 80 basis points will further weaken the Companies' credit metrics.

The GFP was established in 1991 to address the sensitivity of the DCF model to interest rate fluctuations and to address the unreasonably low returns that are produced by this methodology when interest rates are low. The interest rate environment today is certainly much lower than in 1991 when the Commission decided that the DCF model was not producing reasonable results. While the RD specified conventions for the DCF and CAPM in the GFP, the RD in that case left open the opportunity to consider other methods for "good reason."

The results of Staff's DCF model in this case provide that good reason. Staff's application of the DCF model results in an ROE estimate of 8.22 percent, and its CAPM results in an ROE estimate of 8.04 percent. These estimates are 48 to 66 basis points below the lowest ROEs authorized by the Commission in the past 35 years and 142 to 160 basis points lower than the average authorized ROE for gas distribution companies in all U.S. jurisdictions since January 2017. The assumptions and inputs used to develop Staff's DCF and CAPM models do not result in reasonable estimates of the expected return on equity. Benchmarking Staff's DCF and

Moody's takes action on certain National Grid subsidiaries and affirms other ratings, March 29, 2019.

CAPM results against other observable authorized ROEs supports the position that the parameters of the ROE estimation models in this proceeding must be modified to reflect current market conditions. My rebuttal testimony demonstrates that reasonable adjustments to the models result in returns, using Staff's methodology, and the equal weighting proposed in my Direct Testimony, that are reasonable and consistent with the returns provided on investments of comparable risk.

The GFP provided a framework for the Commission to evaluate whether the financial models used to estimate an appropriate ROE were producing reasonable results under a given set of market conditions. The Commission has since considered its methodology and has made modifications to its preferred approach and assumptions to more reasonably reflect market conditions. Specifically, the Commission has modified the averaging periods relied on for stock prices in the DCF model and the bond yields used in the CAPM from six months to three months to more appropriately reflect market data. In addition, the Commission has changed how it estimates the market risk premium from relying on historical data to projected data, abandoning stale data in favor of data that more accurately reflects market conditions that are relevant in setting the ROE. Based on recent market conditions, that have been acknowledged by other regulatory commissions as anomalous, it would be reasonable for the Commission to again evaluate the relevant market data points to be used in the ROE estimation

methodologies to best estimate the forward-looking cost of equity. For the same reasons, it would be reasonable for the Commission to reconsider the weighting placed on the DCF and CAPM approaches to establish a forward-looking ROE that meets the comparability standards of *Hope* and *Bluefield*.

Q. What are your key conclusions and recommendations regarding the appropriate ROE in this proceeding?

A. My key conclusions are as follows:

- Staff's recommended reduction of the Companies' ROE will likely be viewed as credit negative and may result in further downgrades by Moody's Investor Services.
- Staff's reliance on a proxy group comprised of "electric utility proxy companies" is inconsistent with Staff's prior proxy group screening methodology in other cases and results in a return that does not reflect the business and financial risks of the Companies.
- Staff's mechanical application of Beta coefficients in its CAPM analysis suggests a 120-basis point decline in the cost of equity since the Companies' last cases. Such a significant change warrants further analysis, especially given Staff's DCF analysis suggests there has been no change in the cost of equity since the Companies' last case.[7] Yet Staff provides no further analysis or explanation. As

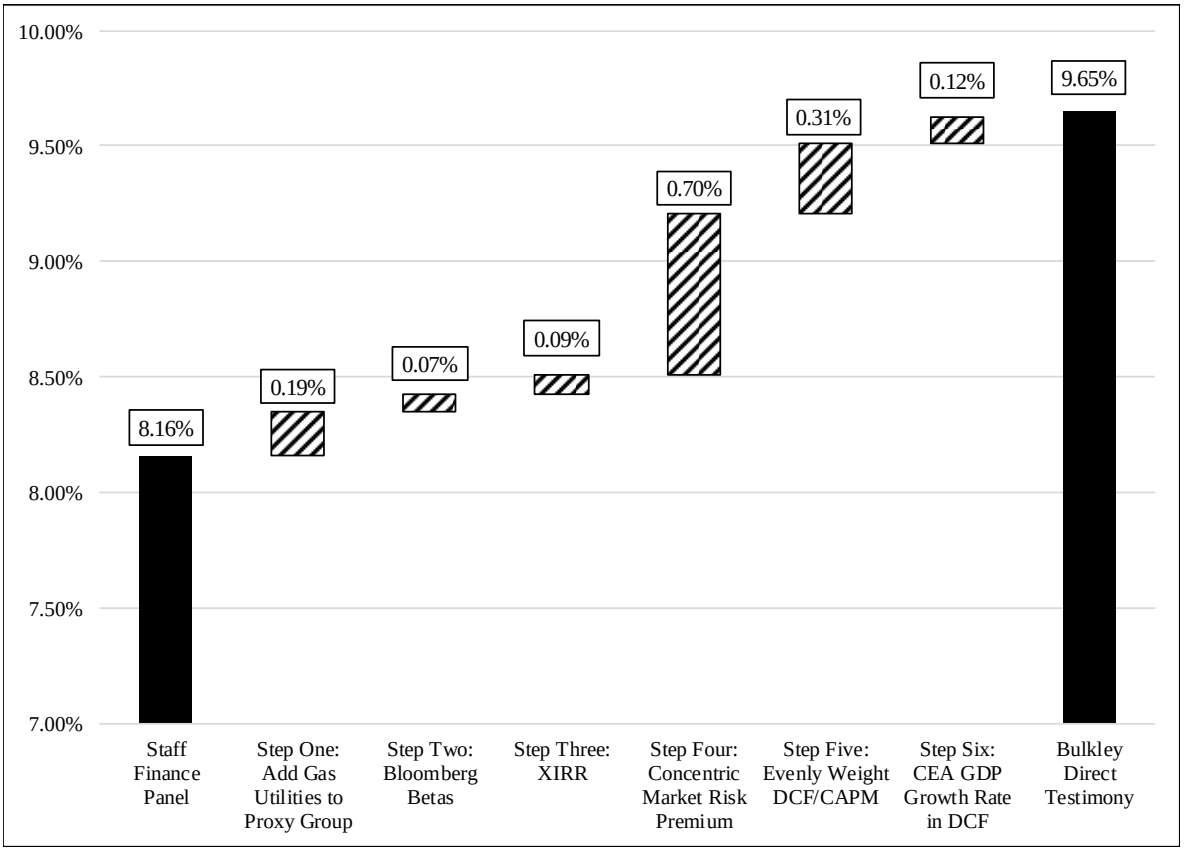
[7] In this case, Staff's DCF result is 8.22 percent. In 16-G-0058 & 16-G-0059 Staff's DCF result was also 8.22 percent.

I described in my Direct Testimony, the performance of utility stocks deviated substantially from the performance of the broader market following the passage of the Tax Cuts and Jobs Act ("TCJA"), while volatility remained high. Staff does not address this fundamental inconsistency and instead relies on analytical results that are below any authorized ROE for a gas distribution company in the U.S. since at least 1980.

- Staff's resistance to use more precise calculation methods that were not available at the time of the GFP is misplaced. As Microsoft Excel has provided advancements that allow for more precise and accurate estimations through the use of the XIRR function, it should be applied in favor of the outdated IRR function.

Section III of my Rebuttal Testimony addresses the technical differences between the analysis presented by Staff and my analysis and recommendation. Reasonable modifications to Staff's proposed assumptions result in significant changes in the resulting ROE. Figure 2 summarizes the effects of modifications made to Staff's DCF model.

Figure 2: Summary of Staff's Adjusted Results[8]



III. RESPONSE TO THE STAFF FINANCE PANEL

Q. What is Staff's ROE recommendation in this proceeding?

A. Staff recommends an ROE of 8.20 percent, based on a two-thirds weighting of its DCF model results of 8.22 percent and a one-third weighting of its average CAPM results of 8.04 percent.[9]

[8] *See* Exhibit ___ (AEB-1R).

[9] Prepared Testimony of the Staff Finance Panel, at 51.

Q. What are the primary areas of disagreement between Staff and you as it relates to the authorized ROE for the Companies?

A. The primary area of disagreement between my analysis and recommendation and that of the Staff is that Staff has dogmatically developed the assumptions used in their ROE formula without considering the effect of market conditions on the models, or the current financial profile of the Companies. As a result, Staff simply runs the models and recommends the results that the model generates. As will be discussed in my testimony, the results of the Staff's model results and ultimate recommendations are not consistent with recently authorized returns in any other jurisdiction and are inconsistent with what the Companies need to maintain their credit ratings, which were downgraded by Moody's earlier this year.

Staff's analysis and criticism of my ROE estimation methodologies are both based on the principle of consistency with prior Commission methodologies. Staff refers multiple times to Commission precedent as the basis for key assumptions in its ROE estimation methodology. The foundation of Staff's criticisms of my methodology is simply that the Commission has not developed the ROE estimation models using the data and methods that I relied on. Staff does not provide any evidence to demonstrate that my assumptions and methods are not those used by investors or are not reasonable. Simply, Staff's position is that my approach

is not what was used in the past. While Staff cites to many instances where it has followed "precedent" with respect to the development of assumptions, Staff fails to acknowledge that there are several instances where the Commission has changed its approach to be responsive to market conditions. This forms the basis of the disagreement between my ROE estimation approach and Staff's methodology.

The specific areas of disagreement related to the calculation of the ROE estimation models are as follows: (1) the composition of the proxy group and the screening criteria used to develop a risk comparable group; (2) the weighting of the DCF and CAPM results; (3) the application of the DCF model and the reasonableness of the results produced by the DCF model under current market conditions; and (4) the application of the CAPM and the reasonableness of making adjustments to the inputs and assumptions used in that model given the current low interest rate environment. The following sections address each of these areas of disagreement.

A. PROXY GROUP COMPOSITION

Q. What is your position with respect to the proxy group composition?

A. An appropriately constructed proxy group consists of companies that are comparable in business and financial risk to the Companies. The importance of selecting a proxy group that is similar in overall financial and business risk to the subject company was endorsed by the United States Court of Appeals for the District of Columbia (the "DC Circuit") in the

Petal Gas Storage decision. The DC Circuit acknowledged that the goal of a proxy group is to rely on companies that possess similar risk to the subject company for the determination of the cost of equity:

> That proxy group arrangements must be risk-appropriate is the common theme in each argument. The principle is well-established. *See Hope Natural Gas Co.*, 320 U.S. at 603 ("[T]he return to the equity owner should be commensurate with returns on investments in other enterprises having corresponding risks."); *CAPP I*, 254 F.3d at 293 ("[A] utility must offer a risk-adjusted expected rate of return sufficient to attract investors."). The principle captures what proxy groups do, namely, provide market-determined stock and dividend figures from public companies comparable to a target company for which those figures are unavailable. *CAPP I*, 254 F.3d at 293–94. Market-determined stock figures reflect a company's risk level and, when combined with dividend values, permit calculation of the "risk-adjusted expected rate of return sufficient to attract investors."
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> What matters is that the overall proxy group arrangement makes sense in terms of relative risk and, even more importantly, in terms of the statutory command to set "just and reasonable" rates, 15 U.S.C. § 717c, that are "commensurate with returns on investments in other enterprises having corresponding risks" and "sufficient to assure confidence in the financial integrity of the enterprise . . . [and] maintain its credit and . . . attract capital," Hope Natural Gas Co., 320 U.S. at 603.[10]

Consistent with the DC Circuit's decision, I have selected a proxy group of companies with comparable investment risk to the Companies. In contrast,

[10] *Petal Gas Storage v. FERC*, 496 F.3d 695, 699-700 (D.C. Cir. 2007).

Staff applied less stringent screening criteria, which result in a larger, but less comparable proxy group.

Q. Do you believe that Staff's proxy group of only electric utilities is appropriate for purposes of setting the authorized ROE for a gas distribution company?

A. No, I do not. Staff has not included any natural gas distribution-only companies in its proxy group, even though the return on equity is being set for gas distribution companies. The business and operating risks for electric utilities and gas distribution companies are different. I strongly disagree with Staff's opinion that the authorized ROE for the Companies should be established based on a proxy group consisting entirely of electric utilities and excluding gas distributors. From the perspective of investors, there would be no reason to dismiss market data for gas distribution companies when setting the return requirements for a gas distribution company. Rather, in my view, investors would give more weight to market data for companies in the gas distribution industry and less weight to electric utilities.

Q. How would the inclusion of the seven gas distribution companies shown in Figure 8 of your Direct Testimony affect the results of Staff's DCF analysis?

A. As shown in Exhibit___(AEB-1R), adding the seven gas distribution companies from my Direct Testimony to Staff's Exhibit__(SFP-18) produces a mean DCF result of 8.47 percent as compared to the 8.22 percent set forth on Exhibit_(SFP-18).

Q. What justification does Staff offer for relying on a proxy group comprised entirely of electric utilities?

A. Staff contends that my use of a Natural Gas proxy group is not appropriate because the Companies' parent is principally viewed by investors as an electric utility holding company.[11] Staff also argues that its electric utility proxy group should be used because the majority of the holdings companies in the group "have operating subsidiaries that are combination electric and gas utilities" and therefore "reflect the challenges and opportunities facing energy utilities as a whole."[12] This ignores the fact that the Companies are not combination electric and gas utilities, but rather are natural gas-only companies. As such, Staff's reliance on electric utilities, and exclusion of gas-only companies, is inappropriate. Staff further argues a proxy group comprised only of natural gas distribution utilities would not produce a

[11] Prepared Testimony of the Staff Finance Panel, at 53-54.

[12] *Id*., at 54.

sufficiently large proxy group, allowing the analysis to be overly influenced by outlier results.[13] Staff does not, however, explain why natural gas distribution utilities should be excluded from the proxy group.

Q. Do investors view the Companies' parent as an electric utility holding company?

A. No, they do not. As an initial matter, Staff does not appear to have offered any evidence demonstrating that the Companies' parent is viewed by investors as an electric utility holding company. As shown in Exhibit ___ (AEB-2R), approximately 45.00 percent of National Grid USA's regulated operating income is derived from gas operations, and approximately 47.00 percent of National Grid USA's regulated revenue is derived from regulated natural gas operations. In contrast, on average, only 12.00 percent of Staff's proxy group's regulated operating income and regulated revenue is derived from regulated natural gas operations. In fact, more than one-third of Staff's proxy group companies do not have any natural gas distribution operations.

Further, in a credit rating report for the Companies' affiliate, the Massachusetts Electric Company, Moody's characterized the National Grid Plc as "well-diversified," with approximately half of National Grid's US rate base comprised of regulated gas utility investments as of March 31,

[13] *Id*., at 80.

2019.[14] This stands in stark contrast to Staff's testimony that KEDNY and KEDLI's "ultimate parent is largely considered to be an electric utility by investors."[15]

Q. Would your opinion regarding the appropriate return on equity for KEDNY and KEDLI change if the Companies' parent was viewed as an electric utility holding company?

A. No, it would not. Even if the Companies' parent were viewed by investors as an electric utility holding company, that would not be relevant when determining the appropriate return on equity for KEDNY and KEDLI. As part of the Company's merger with National Grid in 2006, extensive ring-fencing provisions were imposed on the Companies. These ring-fencing provisions insulated the Companies from the operations of their corporate parent. For example, the Companies are required to maintain (1) an investment-grade credit rating, (2) a debt to capitalization ratio of less than 56.00 percent (for KEDNY) and 58.00 percent (for KEDLI), and (3) a special preferred share that reduces the likelihood of bankruptcy in a distressed situation.[16] Staff themselves concluded that "the rating agencies still view KEDNY and KEDLI as adequately ring-fenced and thus rate the

[14] Moody's Investors Service Credit Opinion, Massachusetts Electric Company, June 14, 2019.

[15] Prepared Testimony of the Staff Finance Panel, at 54.

[16] *See* Exhibit ___ (SFP-5), at 6, and Exhibit ___ (SFP-6), at 4.

Companies' credit quality substantially on a stand-alone basis, apart from their holding Companies."[17]

The ring-fencing provisions limit the risks to which the Companies are exposed to the parent company's risk, and therefore the Companies' return on equity should be established based on the risk profile of the Companies, and not the risk profile of the parent company. Staff and I agree that debt investors view the Companies substantially on a stand-alone basis because they are insulated from risks associated with National Grid. For example, when summarizing the credit quality of both KEDNY and KEDLI, Moody's begins by discussing the risks associated with gas distribution operations, not electric utility operations.[18] Thus, the Companies' return on equity should be established without considering the operations of the corporate parent.

Given the screening criteria applied by Staff, it is not apparent that there would be any change to the proxy group selection for a vertically integrated electric utility. However, as described in more detail below, the risk profile for a vertically integrated electric utility is meaningfully different from a natural gas distribution company. That demonstrates that Staff has not

[17] Prepared Testimony of the Staff Finance Panel, at 21.

[18] Exhibit ___ (SFP-5), at 1, and Exhibit (SFP-6), at 1.

properly accounted for the risks faced by the Companies, which operate exclusively as natural gas distribution utilities.

Q. Is Staff's testimony in this case regarding the appropriate proxy group consistent with its testimony in other cases?

A. No. In 2016, the Staff Finance Panel offered extensive testimony regarding the appropriate cost of capital for National Fuel Gas Distribution Corp ("NFG") in Case 16-G-0257. Despite the fact that NFG's operations were not as clearly distinct from its parent as the Companies (*e.g.* NFG does not issue its own debt or equity),[19] Staff recognized the importance of including natural gas companies in the proxy group for the gas distribution utility, stating that (emphasis added):

> *The starting point for establishing the proxy group is confirming that all of the companies in the group are in the same specific industry classification as the target company. This is done in order to identify each company as a true peer within that proxy group.* Once the appropriate group of peer companies is established, careful consideration must be given to determine the appropriate screening criteria in order to obtain a group of companies that is large enough without becoming unwieldy. Our objective is to select a representative group of companies whose risks closely match those of [NFG]. A careful balance must be struck between these two potentially conflicting goals. While the objective is to select a group of companies whose risks closely match those of the company being examined, it is also large enough in order that we may have sufficient confidence in its results.[20]

[19] Case 16-G-0257, Prepared Testimony of the Staff Finance Panel, at 14.

[20] Case 16-G-0257, Prepared Testimony of the Staff Finance Panel, at 45-46 [clarification added].

Staff's testimony continued (emphasis added):

> *Ideally, we would have selected a proxy group comprised entirely of regulated gas utilities whose risk profile represents, as closely as possible, the risk characteristics of the individual company being examined.* However, given that the entire Value Line Investment Survey (Value Line) universe of gas utility companies consists of an insufficient number of publicly traded water companies upon which to base our proxy group to produce a reasonable result, we expanded our proxy group to include electric utilities within the Value Line universe that face similar risks and of which investors have similar expectations.[21]

In 2016, the Staff Finance Panel offered testimony regarding the appropriate cost of capital for the Corning Natural Gas Corporation ("Corning") that was almost identical to the testimony quoted above from the NFG case.[22]

Q. What implications does Staff's testimony in the NFG and Corning cases have for KEDNY and KEDLI's cost of capital?

A. As described above, adding the seven gas distribution companies from my Direct Testimony to Staff's Exhibit__(SFP-18) produces a mean DCF result of 8.47 percent as compared to Staff's mean DCF result of 8.22 percent. As shown in Exhibit___(AEB-1R), if the DCF analysis was limited to only the seven gas distribution companies from my Direct Testimony, the mean DCF result would be 9.48 percent. Therefore, Staff has understated its DCF

[21] *Id*., at 46-47.

[22] Case 16-G-0369, Prepared Testimony of the Staff Finance Panel, at 45-47.

results by 126 basis points by failing to select "a proxy group comprised entirely of regulated gas utilities whose risk profile represents, as closely as possible, the risk characteristics of the individual company being examined," consistent with Staff's prior position.[23]

Q. Do you share Staff's concern that there are only seven companies in the Natural Gas Proxy Group?

A. No, I do not. As described in my Direct Testimony, the primary objective of the screening criteria used to select a proxy group is to ensure that the resulting utilities have risk profiles that are as comparable as possible to the Companies.[24] A larger proxy group is not inherently preferable to a smaller proxy group if the larger proxy group is less comparable to the Companies. While I recognize that a proxy group of seven utilities is somewhat limited relative to the Combined Utility Proxy Group, in the NFG and Corning Gas cases, Staff acknowledged that the business and financial risks faced by the Natural Gas Proxy Group are more comparable to these natural gas distribution companies. Considering KEDNY and KEDLI on a stand-alone basis, the natural gas proxy group is also more comparable to the Companies than the Combined Utility Proxy Group. Further, as illustrated in Figure 14 of my Direct Testimony, the average DCF/CAPM results of the Natural Gas Proxy Group are corroborated by the DCF/CAPM results of the Combined

[23] Case 16-G-0257, Prepared Testimony of the Staff Finance Panel, at 46-47.
[24] Direct Testimony of Ann E. Bulkley, at 40.

Utility Proxy Group. This demonstrates that the Natural Gas Proxy Group provides a reasonable indication of the Companies' cost of equity.

Q. What additional concerns does the Staff Finance Panel have with the selection criteria you utilized to select your proxy group?

A. Staff has three additional concerns with my proxy group selection criteria. First, Staff argues that I should have screened based on the percentage of revenue derived from regulated utility operations, rather than screening based on the percentage of operating income derived from regulated utility operations. Second, Staff claims that I misapplied a screening criterion that each company must be covered by more than one equity analyst. Third, Staff believes that I should not have limited my proxy group to transmission and distribution only utilities by excluding companies with owned generation comprising more than 60.00 percent of MWh sales to ultimate customers.

Q. Why is net operating income, rather than revenue, the appropriate screening criterion to utilize for selecting a proxy group for the Companies?

A. As discussed extensively in my Direct Testimony,[25] and expanded upon in response to Information Request DPS 894 (*see* Exhibit__ (AEB-3R)), net operating income better reflects the contribution of each business segment

[25] *Id.*, at 43-45.

to the corporation's earnings. For regulated utilities that collect the cost of purchased gas, fuel, and/or power on a pass-through basis through rates, regulated revenue can fluctuate materially with changes in weather without impacting the corporation's earnings or financial position. In contrast, net operating income, which excludes purchased commodity costs, more closely represents the effect of each business segment on the corporation's overall risk profile. Additionally, as discussed further in response to Information Request DPS-894, multiple other regulatory agencies have accepted the use of an income-based screening criterion.

Q. Why does Staff suggest that revenue is the superior screening criterion?

A. Staff argues that using net operating income, rather than revenue, would inappropriately exclude comparable companies from the proxy group.[26] To illustrate this point, Staff observes that the revenue criterion would result in the exclusion of NextEra Energy, Inc. ("NextEra") from the proxy group. Staff further argues that the revenue criterion better accounts for exposure to risky competitive operations than the net income screening criterion.

Q. Is NextEra comparable to the Companies?

A. No. NextEra, as a vertically integrated electric utility, is materially affected by risks associated with its electric generating portfolio. Consider the

[26] Prepared Testimony of the Staff Finance Panel, at 83-85.

discussion of NextEra's financial data offered by Value Line (*i.e.*, the source that Staff relies on extensively in both its DCF and CAPM analyses):

> This should be another good year for NextEra Energy. The company's utility subsidiary, Florida Power & Light, is benefiting from rate relief and healthy customer growth. A $200 million rate increase took effect after a 1,750-megawatt gas-fired plant began operation at the end of March (ahead of schedule) at a cost of $1.2 billion. The utility is also adding solar capacity, which is recovered through a regulatory mechanism. On the nonregulated side, NextEra Energy Resources continues to add wind, solar, and energy-storage capacity. This subsidiary is also repowering older wind projects. Even so... We cut our 2019 share-earnings estimate by $0.75. First-quarter profits were hurt by mark-to-market accounting charges (which we include in our presentation, even though NextEra excludes these) and subpar production from the company's wind assets.[27]

As illustrated by the Value Line report quoted above, the risks NextEra faces as a vertically integrated electric utility are quite different from the risks faced by the Companies. Investors analyzing NextEra's stock consider the risks associated with company's extensive fleet of electricity generating resources, the impact of weather on the company's generation, and the company's ability to add renewable generation capacity. None of those are risks faced by the Companies, which operate as natural gas distribution utilities.

[27] Value Line Investment Survey, May 17, 2019.

Q. How does the risk profile of your Combined Utility Proxy Group compare to the risk profile of the Companies?

A. As illustrated in Exhibit ___ (AEB-8), Exhibit ___ (AEB-9), and Exhibit ___ (AEB-10), and as discussed in more detail below, the utilities in my combined utility proxy group are, on average, less risky than the Companies. This contradicts Staff's objection that "it is quite possible that using the 70% of operating income criteria could introduce companies into the proxy group that have a higher inherent risk profile than is proper for a regulated utility proxy group."[28]

Q. Are there utilities that are comparable to the Companies that are excluded by the revenue screening criterion?

A. Yes. As described in my Direct Testimony, New Jersey Resources would have been excluded from the Combined Utilities Proxy Group if revenue were used as the screening criterion rather than net operating income. It is difficult to understand why Staff would prefer to include vertically integrated electric utilities, instead of natural gas distribution utilities, in the proxy group used to establish the ROE for a natural gas distribution utility. That fact is especially difficult to understand in light of the extensive ring-fencing provisions insulating the Companies from the operations of the corporate parent.

[28] Prepared Testimony of the Staff Finance Panel, at 84.

Q. Is Staff correct that you inadvertently eliminated three companies that are covered by multiple industry equity analysts?

A. No, Staff has misunderstood the screening criterion. Staff argues that, because Otter Tail, MGE Energy, Inc, and El Paso Electric are followed by multiple equity analysts, each of those companies should have passed my screening criterion.[29] However, the screening criterion does not test whether each company is followed at all by more than one equity analyst. Rather, the screening criterion tests whether the earnings per share growth rates provided by First Call and/or Zacks reflect consensus analyst estimates. As discussed in more detail later in my testimony, consensus estimates that reflect the independent opinions of multiple analysts are more reliable than estimates developed by a single analyst because they are less influenced by idiosyncratic error. The earnings growth rates provided by First Call and Zacks for each of the three utilities identified by Staff do not reflect consensus estimates, and thus each utility was excluded from the Combined Utility Proxy Group. The fact that Staff has identified equity analysts such as the Williams Capital Group and the Maxim Group that cover Otter Tail, MGE Energy, Inc., and El Paso Electric does not mean that there are standardized consensus earnings growth rate estimates available for those utilities.

[29] *Id*., at 85-86.

Q. Is your testimony regarding the importance of relying on consensus estimates similar to Staff's testimony regarding the importance of a sufficiently large proxy group?

A. Yes, it is. Staff explained that relying on a sufficiently large proxy group mitigates the undesirable effects of biased and/or inaccurate growth estimates on the ultimate DCF results.[30] It is important to rely on estimates that reflect a consensus of analysts for that same reason: the more estimates that are reflected, the less the results are influenced by outlier growth rates. For example, Staff takes issue with my use of a 25.50 percent earnings growth rate from Value Line for Northwest Natural Gas Company.[31] However, Staff did not point out that the effect of that growth rate on my mean results is moderated because it is averaged with the considerably lower earnings growth rates projected by First Call and Zacks. Had I relied solely on Value Line, as Staff has, the effect of outlier growth rates would be much more substantial.

B. WEIGHTING OF DCF AND CAPM METHODOLOGIES

Q. Why did the Commission initiate the GFP?

A. As discussed in my Direct Testimony, the GFP was initiated because the Commission recognized that the DCF method was particularly sensitive to interest rate fluctuations and was producing returns far below the returns

[30] *Id*., at 52-53.
[31] *Id*., at 89.

produced by other methodologies.[32] The RD proposed that a two-third/one-third weighting be assigned to the results of the DCF and CAPM analyses, respectively, with less weight given to CAPM because it had only been used to that point as a check on the DCF model.[33] While Staff relies on the GFP to support its decision to place two-thirds weight on the DCF results, Staff does not acknowledge that the RD in the GFP left open the possibility that the weightings and methodologies could be adjusted if necessary to ensure that the results promote regulatory credibility.

Q. Does Staff provide examples of the Commission's thoughts regarding the weighting of the DCF and CAPM results?

A. Yes. Staff cites several cases and provides one specific example of the Commission's discussion of the weighting of the DCF and CAPM results from Case 06-E-1433. As noted by Staff, in that case, the Commission continued to place two-thirds weight on the DCF model results and one-third weight on the CAPM results.

[32] Case 91-M-0509, Proceeding on Motion of the Commission to Consider Financial and Regulatory Policies for New York State Utilities, Recommended Decision, (issued July 19, 1994) ("Generic Finance RD"), at 2.

[33] *Id*., at 27.

Q. What are the important factors to consider in reviewing the Commission's decision in Case 06-E-1433?

A. First, it is important to note that the Commission's decision was issued in October 2007, before the financial crisis of 2007-2009 and the ensuing Great Recession, as well as the Federal Reserve's extended involvement in the financial markets that began following those events and continues today. Therefore, the market data used in Case 06-E-1433 to estimate the projected ROE were not influenced by anomalous market conditions such as have been experienced in recent history and that affect the market data used in the ROE estimation methodologies in these proceedings.

Second, the Commission notes that it changed its calculation of the market return used in the estimation of the market risk premium in the CAPM. The Commission recognized that the use of historical returns published by Ibbotson were stale and less reliable and therefore began relying on projected returns as published by Merrill Lynch. In this same case, the Commission recognized that six-month average stock prices could be "stale."[34] Currently, the Commission's methodology relies on three-month average stock prices. These types of changes demonstrate that the Commission is willing to consider modifications to the ROE estimation

[34] Case 06-E-1433, *Orange and Rockland Utilities*, "Order Setting Permanent Rates, Reconciling Overpayments During Temporary Rate Period, and Establishing Disposition of Property Tax Refunds" (October 18, 2007) at 11.

methodology to include more current and relevant information as market conditions change.

Finally, it is important to note that the Commission decision in Case 06-E-1433 did not state that it would never consider changing the weights on the ROE estimation methodologies. Rather, the Commission's conclusion at that time, nearly ten years ago, was that it was "not ***now*** inclined to deviate from our long-held view that the CAPM should not be entitled to more than one-third of the weight."[35] The Commission explicitly left open the possibility that there would be a point in the future when it would be appropriate to consider such a change. Based on the data I have presented, and the viewpoints provided by other regulatory commissions, it is reasonable to conclude that current market conditions now warrant such a change.

Q. Why do you believe that current conditions in financial markets support reconsideration of the weight placed on the DCF and CAPM Methodologies?

A. When the RD was issued in the GFP in 1991, one of the primary concerns identified by the Commission was that the low interest rate environment was causing the DCF model to understate investors' return requirements.[36]

[35] *Id*., at 15. (Emphasis added.)

[36] 1994 N.Y. PUC Lexis 141, *37.

The Commission also noted that there was nothing sacrosanct about the DCF return on equity analysis.[37] The average daily yield on 30-year Treasury bonds in 1991 was 8.14 percent, whereas the average daily yield on 30-year Treasuries in 2019 has been 2.75 percent.[38] The extraordinarily low interest rate environment today should do nothing to alleviate concerns about how the DCF model is affected by the interest rate environment. On the contrary, if the interest rate environment in 1991 was sufficient reason for the RD in the GFP to conclude that placing one-third weight on the CAPM results was appropriate, then the current interest rate environment should provide sufficient basis for a conclusion that the weighting of the DCF and CAPM methodologies should be modified in these cases.

Q. What are your conclusions regarding the weighting of the DCF and CAPM methodologies?

A. The Companies' proposed ROE of 9.65 percent, which is highly consistent with the weighted average 67%/33% DCF/CAPM mean result for the Combined Utility Proxy Group and the 67%/33% DCF/CAPM mean result for the Natural Gas Proxy Group, is appropriate. While I understand the desire to employ a consistent process to estimate the return on equity, as discussed in the *Hope* decision, "[u]nder the statutory standard of 'just and reasonable,' it is the result reached, not the method employed, which is

[37] *Ibid*.

[38] Source: Bloomberg professional, as of, August 30, 2019.

controlling." Staff's ROE recommendation of 8.20 percent, which is based on a weighting of the DCF and CAPM results that was established 25 years ago, is not comparable to returns available to investors in other jurisdictions for companies with comparable risk.

C. APPLICATION OF THE DCF MODEL

Q. Please summarize Staff's application of the DCF model.

A. Staff uses a two-stage DCF model that forecasts dividends from 2019 to 2023 using Value Line's estimates of projected dividends in that period, and a "sustainable growth rate" from 2023 forward. Using the DCF methodology, Staff calculates a median ROE for its proxy group of 8.22 percent. Staff applies a 2/3 weight to its DCF model results in deriving its overall ROE recommendation.

Q. What are the differences between your application of the DCF model and Staff's approach?

A. The most significant difference between my application of the DCF model and Staff's approach is the growth rates we use in our respective analyses. I have used a consensus of analysts' EPS growth rates for the proxy group companies as the near-term growth rate, and an estimate of growth in the overall economy for the long-term growth rate. This mitigates the uncertainty associated with forecasting individual companies' growth rates over very long-time horizons. By contrast, Staff uses dividend growth projections from a single source (*i.e.*, Value Line) for their near-term growth

rate, and a "sustainable growth rate" for their long-term growth rate. Staff states that it is "highly unlikely that investors would rely exclusively on the earnings per share growth rate forecasts of Wall Street analysts in determining short-term dividend projections".[39]

Q. Do you agree with Staff's position on earnings growth rates?

A. No, I do not. Staff prefers dividend growth rates rather than earnings growth rates as the near-term growth rate in their DCF analysis. However, as discussed in my Direct Testimony, dividend growth is fundamentally driven by earnings growth.[40] While the model is, indeed, called the "Discounted Cash Flow" model, the cash flows it refers to are those an investor can expect to receive during the time they own the stock. Those cash flows are quarterly dividend payments plus any capital appreciation that occurs between when the stock is purchased and when it is sold. Dividend payments and capital appreciation are both a function of earnings per share ("EPS"), which is ultimately what determines the return on equity to the investor. As discussed in my Direct Testimony, dividends are based on management decisions related to cash management and other factors, and therefore dividend growth rates are less likely to accurately reflect investors' growth expectations than earnings growth rates.[41]

[39] Prepared Testimony of the Staff Finance Panel, at 90.

[40] Direct Testimony of Ann E. Bulkley, at 58.

[41] *Id.*, at 46-47.

I have relied on earnings growth because earnings are the fundamental determinant of a company's ability to pay dividends. As noted by Brigham and Houston:

> Growth in dividends occurs primarily as a result of growth in earnings per share (EPS). Earnings growth, in turn, results from a number of factors, including (1) inflation, (2) the amount of earnings the company retains and invests, and (3) the rate of return the company earns on its equity (ROE).[42]

Investment analysts report predominant reliance on EPS growth projections. In a survey completed by 297 members of the Association for Investment Management and Research, the majority of respondents ranked earnings as the most important variable in valuing a security (more important than cash flow, dividends, or book value).[43]

Academic research also supports the use of EPS growth estimates. A 2002 study in the *Journal of Accounting Research*, examined "the valuation performance of a comprehensive list of value drivers" and found that "forward earnings explain stock prices remarkably well" and were generally superior to other value drivers analyzed.[44] A 2012 study from the journal *Contemporary Accounting Research* found that the sell-side analysts with

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[42] Eugene F. Brigham and Joel F. Houston, *Fundamentals of Financial Management*, at 317 (Concise Fourth Edition, Thomson South-Western, 2004).

[43] Block, Stanley B., "A Study of Financial Analysts: Practice and Theory", Financial Analysts Journal (July/August 1999).

[44] Liu, Jing, et al., "Equity Valuation Using Multiples," Journal of Accounting Research, Vol. 40 No. 1, March 2002.

the most accurate stock price targets were those whom the researchers found to have more accurate earnings forecasts.[45]

Q. According to Staff, the Value Line growth rates are produced by a group of analysts, and therefore represent a consensus view. Do you agree?

A. No, I do not. Staff provides a letter from Value Line describing the development of its growth projections, as Exhibit ___(SFP-25). Value Line states that each company it covers is assigned to a lead analyst who is responsible for building the coverage model for that company. Specifically, the letter states, "[e]ach stock in The Value Line Investment Survey is assigned to a specific analyst."[46] The letter goes on to report the quality control procedures applied to the analyst's report. Nowhere in the letter, titled "Quality Control Procedures," does Value Line describe a process whereby multiple independent evaluations are performed and then averaged together to form a consensus view. That is the important element missing from Staff's analysis: Staff relies on a source for growth rate estimates that reflects the opinion of a single analyst.

While I agree that Value Line is a trusted source for investment professionals, it is not the only tool that investors rely on to make decisions.

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[45] Gleason, C.A., et al., "Valuation Model Use and the Price Target Performance of Sell-Side Equity Analysts," *Contemporary Accounting Research*.

[46] Exhibit __ (AQ-15).

There are additional data sources readily available that compile the consensus viewpoints of multiple brokerage analysts, and it is reasonable to expect that investors also consider that information. In fact, there have been studies performed comparing Value Line and I/B/E/S analyst earnings forecasts in terms of accuracy, rationality and as proxies for market expectations. In 2001, a study concluded that "I/B/E/S forecasts were superior, as explained by the combination of I/B/E/S's timing advantage and the mitigation of idiosyncratic error through consensus building."[47] Furthermore, the I/B/E/S long-term forecasts were less biased and more accurate.[48]

In addition, Staff's position on the position on the appropriate growth rate is contradictory. Staff criticizes the use of earnings growth rates because "Investors will not just look at expected earnings but also factor in all relevant information when estimating their required return."[49] If investors consider all relevant information, it is inconsistent with that principle to limit the DCF analysis to rely exclusively on a single source of estimated growth rates, as Staff has done. Based on Staff's conclusion, it is reasonable to consider the various sources of consensus growth estimates that I have relied on in addition to the Value Line earnings growth rates.

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47 Ramnath, Sundaresh, Rock, Steven, Shane, Philip, "Value Line and I/B/E/S Earnings Forecasts, November 8, 2001, at 1.

48 *Ibid*.

49 Prepared Testimony of the Staff Finance Panel, at 90.

Q. Do you agree with Staff that reliance on Value Line growth rates lends a degree of predictability to the New York rate setting process for the investment community?

A. No, I do not. The investment community reviews the returns established through the New York rate setting process based on the reasonableness and comparability of the returns to other investment alternatives.

Q. Please summarize your concerns with Staff's sustainable growth rate.

A. I have two main concerns with Staff's sustainable growth rate. First, while Staff asserts that its calculation of the sustainable growth rate represents a measure of long-term growth for the period 2024 and beyond, the inputs to its calculation are, for the most part, shorter-term estimates for the period 2019 and earlier. For example, the "b*r" component of Staff's sustainable growth rate is derived from Value Line forecasts that only extend through 2022. Therefore, Staff's long-term growth rate estimate reflects, at best, one analyst's forecast of only the very early years of the second stage of Staff's DCF model, which theoretically extends into perpetuity. In contrast, my estimate of long-term GDP growth reflects projections over 30 years or more and considers overall measures of economic growth. [50] Second, Staff's sustainable growth rate relies on Value Line's estimate of each proxy company's ROE, as the "r" in the "b * r" component of Staff's growth rate

[50] Direct Testimony of Ann E. Bulkley, at 62-65.

is the expected ROE. This introduces an element of circularity into Staff's calculation, that it fails to reconcile with its results. That is, based on Exhibit __ (SFP-17), the mean and median ROEs assumed in Staff's DCF calculation are 11.14 percent and 10.64 percent, respectively, yet Staff's analysis suggests the proxy group average ROE is 8.22 percent. By that measure, the range of ROEs established in my analysis for my proxy group companies (*i.e.*, 9.65 percent to 10.55 percent) is quite conservative relative to the proxy group ROEs assumed in Staff's sustainable growth estimate.[51]

Q. Have other regulatory commissions abandoned the use of the sustainable growth rate?

A. Yes. In Opinion No. 531, FERC noted that the sustainable "br + sv" growth estimate produces a projection of short-term growth, similar to the IBES growth projections. FERC determined that the use of the IBES projections, without the sustainable growth rate were appropriate in a two-stage DCF model.[52] FERC made no modification to this approach in its recently proposed methodology.[53]

[51] *Id*., at 67.

[52] FERC Opinion 531, at 34, 39.

[53] FERC, Docket No. EL11-66-001, et al., Order Directing Briefs, issued October 16, 2018.

Q. Why is it reasonable to rely on historical averages of real GDP growth to approximate future economic activity?

A. I disagree with Staff that historical averages "are poor indicators of future economic activity."[54] Based on current and recent market conditions, the use of historical real GDP growth is more appropriate than using a current projection of real GDP growth. Economists have reviewed historical growth patterns related to severe financial crises and have concluded that estimates of GDP growth have generally been understated in the decade following severe financial crises. Specifically, the financial crisis and recession that began in 2007 were qualitatively different from most other U.S. economic downturns, which were followed by a rapid return to pre-recession overall output growth levels. In that regard, the current U.S. economic growth situation is similar to that following the two most severe economic events in U.S. history (*i.e.*, the 1929 stock market crash and the 1973 oil shock). Economists who have examined the repercussions of those two historical crises (and similar severe financial crises in other countries) have found that GDP growth rates tended to be lower during the decade following such events.[55] Therefore, it would not be appropriate to assume that current projections of GDP growth are representative of long-term GDP growth starting in 2029 and continuing for the next 200 years.

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[54] Prepared Testimony of the Staff Finance Panel, at 95.

[55] *See*, Reinhart, Carmen M. and Vincent R. Reinhart, "After the Fall," NBER Working Paper 16334, September 2010, in Federal Reserve Bank of Kansas City Economic Policy Symposium Volume, *Macroeconomic Challenges: The Decade Ahead* at Jackson Hole, Wyoming, on August 26-28, 2010, at 2.

Q. Have you performed an analysis of historical GDP growth rates?

A. Yes. I compared the average real GDP growth in the first ten years following the two historical economic crises most comparable to the recent financial crisis (*i.e.*, the 1929 stock market crash and the 1973 oil shock) to the average real GDP growth in the next two decades following each crisis (*i.e.*, eleven to 30 years after the events). I did the same for each of the twentieth-century U.S. recessions for which sufficient data are available. My findings are presented in Figure 3.

Figure 3: Real GDP Growth Rates Following U.S. Economic Downturns[56]

Event	Compound Average Real GDP Growth Rate		
	Decade Following Crisis	Next Two Decades	Difference (Basis Points)
Major Economic Crises			
1929 Stock Market Crash	2.06%	4.72%	266
1973 Oil Shock	2.55%	3.39%	83
Other Recessions			
1937	6.68%	4.15%	-253
1945	3.77%	3.59%	-18
1948	3.79%	3.95%	16
1953	3.60%	3.23%	-37
1957	4.84%	3.13%	-170
1960	4.41%	3.28%	-112
1969	3.57%	3.01%	-56
1980	3.32%	2.45%	-88
1981	3.52%	2.62%	-90

[56] Real GDP data are from the U.S. Bureau of Economic Analysis. The years in which each recession started are from the National Bureau of Economic Research ("NBER"), "US Business Cycle Expansions and Contractions," available at

Figure 3 shows that real GDP growth in the first ten years following the 1929 stock market crash and the 1973 oil shock was substantially lower than real GDP growth in the next two decades following each event. In contrast, eight out of the nine other twentieth century U.S. economic downturns analyzed showed the opposite pattern. In light of the academic research cited above and the findings presented in Figure 3, it is reasonable to believe that current projections of real GDP growth are understated. For that reason, the most reasonable means to forecast long-term GDP growth is to assume a return to long-term historical rates of real GDP growth and to estimate long-term nominal GDP growth based largely on market-based, long-term inflation estimates.

Q. Has Staff relied on historical data in any of their analyses?

A. Yes. For example, in its CAPM analysis, Staff relies exclusively on Value Line Betas that use a 5-year historical estimate of the relative risk of proxy companies and the overall market.

Q. Do you have any additional concerns with the results of Staff's DCF analysis?

A. Yes. One quarter of Staff's proxy group (*i.e.*, seven of the twenty-eight utilities) has a DCF result below 7.00 percent. According to Staff, the risk-

http://www.nber.org/cycles.html. Note that this table excludes the three most recent recessions, which started in 1990, 2001, and 2007 owing to a lack of sufficient data for GDP growth in the following years to calculate comparable long-term GDP growth rates.

free rate as of June 2019 is 2.55 percent.[57] The implied equity risk premium for the proxy group is calculated as the difference between the risk-free rate and the ROE results produced by the two versions of the CAPM developed by Staff. As shown in Exhibit ___ (SFP-20), Staff's traditional CAPM ROE result is 7.63 percent, which implies an equity risk premium of 5.08 percent. Staff's Zero Beta CAPM ROE result is 8.44 percent, which implies an equity risk premium of 5.89 percent. The CAPM analysis provided with my Direct Testimony implies equity risk premiums from 6.37 percent to 8.31 percent, depending on which inputs are reflected.

However, an ROE result of 7.00 percent implies an equity risk premium of 4.45 percent. That is well below the equity risk premiums implied by my analysis, and by Staff's analysis. The fact that one quarter of the utilities in Staff's proxy groups have unreasonably low DCF results calls into question the reliability of that model.

Q. Staff recommends that the Commission reject your use of the "XIRR" function in favor of the "IRR" function. What is Staff's basis for that recommendation?

A. Staff contends that my DCF analysis inappropriately relies on an extended version of Excel's internal rate of return formula. Specifically, Staff takes issue with my use of Microsoft Excel's XIRR function, which can take into account the timing of quarterly dividend payments. Rather, Staff argues in

[57] Exhibit ___ (SFP-20).

favor of the imprecise IRR function, which cannot take into account the timing of quarterly dividend payments when applied to annualized calculations. Staff's basis for rejecting the XIRR function is that (1) the IRR function is suitable for regular cash flows, and (2) the Commission has historically relied on the IRR function.[58]

Q. Please explain the difference between the IRR and XIRR functions.

A. As described in more detail in my response to Information Request DPS-895 (*see* Exhibit__ (AEB-3R)), both the IRR and XIRR functions are Microsoft Excel functions that are used to calculate the internal rate of return for a series of cash flows. The IRR function makes the simplifying assumption that cash flows are available at a set time in each period. In contrast, the XIRR function allows additional inputs as to the timing of cash flows. Specifically, the XIRR function allows the analysis to reflect the assumption that dividends are paid, on average, halfway through the year (*i.e.*, two dividends are paid in the first half of the year, and two dividends are paid in the second half of the year). Staff has provided an analysis demonstrating that the difference in the underlying assumptions between the IRR and XIRR formulas could be as much as 70 basis points.[59] In this case, the difference in the underlying assumptions between the IRR and XIRR formulas has an effect of 11 to 21 basis points on my DCF analysis.

[58] Prepared Testimony of the Staff Finance Panel, at 97-100.

[59] *Id*., at 98.

Q. Why is the XIRR function preferable to the IRR function?

A. The XIRR function allows the analysis to reflect a more nuanced set of assumptions regarding the timing of cash flows. If Staff were correct that the IRR function adequately reflected the timing of cash flows available to utility investors, then the IRR and XIRR functions should return consistent results. The fact that there is a material difference in the results produced by both functions illustrates that Staff's assessment is incorrect.

Q. Please provide an example of why the XIRR function is appropriate, even for regular cash flows.

A. Suppose that there are two dividend paying utilities, identical in every respect but one: utility one, pays a lump-sum dividend once per year; and utility two, pays daily dividends that are equal, on an annualized basis, to the lump-sum dividend paid by utility one. According to Staff's analysis, the result of a discounted cash flow analysis of both utilities would be identical. However, it is a well-recognized financial principle that investors place value the timing in which cash flows are received. While both utilities in this hypothetical pay "regular" dividends, the timing of cash flows matters. The XIRR function allows the analysis to better reflect the timing of cash flows. In contrast, the oversimplified and imprecise IRR function assumes that all dividends are paid at the end of the year. Every utility in my proxy groups, and in Staff's proxy group, pays dividends on a quarterly

basis. Thus, the IRR function, which assumes that investors have to wait longer to receive dividend payments from the utility stock, understates the cost of equity.

Q. When was the XIRR function added to Excel?

A. The XIRR function was not fully integrated into Microsoft Excel until 2007, or more than a decade after the Generic Finance Proceeding ("GFP").[60] As explained in my Direct Testimony, my analysis extends the principles advanced in the Recommended Decision ("RD") in the GFP to reflect best practices in financial analysis today. The XIRR function, which the RD in the GFP would not have contemplated since it was not readily available at the time, is a clear example of a best practice in financial analysis.

Q. What do you conclude about the results of the DCF models under current market conditions?

A. As explained in my Direct Testimony, the DCF models are not producing reasonable results under current market conditions as a result of low dividend yields and high stock valuations.[61] High valuations on utility shares, as suggested by Value Line, could result in an underestimation of the cost of equity using the DCF model, especially if those high valuations are not sustainable in the future. For these reasons, I believe it is appropriate to afford an appropriate weight to the results of other ROE estimation

[60] *See* Microsoft, https://blogs.msdn.microsoft.com/eric_carter/2009/03/19/object-model-changes-for-developers-between-excel-2003-and-excel-2007.

[61] Direct Testimony of Ann E. Bulkley, at 31-32.

methodologies. In particular, the CAPM is more sensitive to changes in interest rates, which is an important risk factor to investors in the current financial market.

D. Application of the CAPM

Q. Please summarize Staff's application of the CAPM.

A. Staff developed its estimate of the risk-free rate of 2.55 percent by using a recent average of 10-year and 30-year Treasury bond yields. For the Beta estimate, Staff used the average Value Line Beta for its proxy group of 0.61. To estimate the market risk premium, Staff subtracted its risk-free rate estimate from an average of the forecast returns for the S&P 500 from Merrill Lynch's May, June and July 2019 Quantitative Profiles. Staff also developed a zero-Beta CAPM using similar inputs as its traditional CAPM analysis, but using a market risk premium based on weighting the Beta times the market risk premium by 0.75 and the market risk premium itself by 0.25. Staff relied on the average of its two CAPM analyses and weighted that result by one-third in the formulation of its overall ROE recommendation.

Q. What are the primary differences between your application of the CAPM and Staff's application of this model?

A. Because the estimation of the ROE is a forward-looking concept, and the ROE that is authorized in these cases will be in effect for some period in the future, my analysis appropriately considers both the recent historical risk-free rate and the projected risk-free rate. The analysis presented in my

Direct Testimony also relies on Bloomberg estimates of Beta that more accurately reflect market conditions. Finally, I estimate the market risk premium based on the difference between the return on large company stocks, as measured by the S&P 500, and the yield on 30-year Treasury bonds.

Q. Why is the use of 30-year Treasury bond yields more appropriate than relying on 10-year and 30-year Treasury bond yields?

A. Staff's rationale for relying on 10-year and 30-year Treasury bond yields in its CAPM analysis is that it is consistent with the approach the Commission has relied on in prior cases, and the use of these securities reflects the expectations of investors who have "intermediate as well as long-term investment horizons.[62] I disagree with Staff's rationale. As noted by Morningstar, "the horizon is a function of the investment, *not the investor*." The use of the 30-year Treasury bond yield as the risk-free rate is consistent with the investment horizon for electric and natural gas utility assets. Taking Staff's rationale to the limits, an investor who plans to hold a position in a utility equity share for only six months would use a certificate of deposit rate to evaluate the potential return. That is clearly not the case for any rational investor considering return requirements.

[62] Prepared Testimony of the Staff Finance Panel, at 114.

Q. Have other regulatory agencies commented on the appropriate securities to use for the risk-free rate in the CAPM?

A. Yes. In Opinion No. 531, the FERC relies on the yield on the 30-year Treasury bond.

> As noted above, the 30-year U.S. Treasury bond yields are a generally accepted proxy for the risk-free rate in a CAPM analysis, and are also considered superior to short- and intermediate-term bonds for this purpose.[63]

FERC made no modification to this approach in its recently proposed ROE methodology.[64]

Q. Why are projected treasury bond yields informative in the CAPM analysis?

A. As discussed in my Direct Testimony, recent market conditions, particularly recent artificially low interest rates, have had an effect on the results of the ROE estimation models. Given these market conditions, I disagree with Staff's position that "current rates are the best indicator of future rates, as they are based on the latest available information to investors."[65] Just as investors have available to them actual and projected growth rates for individual companies, they also have available current and projected yields on Treasury bonds.

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[63] FERC Order 531-B at para 114. Citing also to Roger A. Morin, New Regulatory Finance 151-152 (Public Utility Reports, Inc. 2006) ('the yield on very long-term government bonds, namely, the yield on 30-year Treasury bonds, is the best measure of the risk-free rate for use in the CAPM and Risk Premium methods.").

[64] FERC, Docket No. EL11-66-001, et al., Order Directing Briefs, issued October 16, 2018.

[65] Prepared Testimony of the Staff Finance Panel, at 114.

Furthermore, the use of projected Treasury bond yields is consistent with the use of a forward-looking market risk premium. For this same reason, it is not appropriate to rely only on 3-month average Treasury bond yields, when investors and the market generally are expecting interest rates to increase from the historical lows that have recently been experienced in the market. This is particularly important when the expectation is that interest rates will increase over the period that the ROE that is determined in these cases will be in effect. As discussed previously, consensus forecasts predict interest rates to increase in the near and longer-term. It is reasonable to expect that investors can and do consider this information, as they do other market projections. Therefore, it is reasonable, and consistent with Staff's position regarding the calculation of the market risk premium, to rely on expectations of market conditions to the extent that those expected conditions differ from recent history.

Finally, from a practical perspective, the ROE that is approved by the Commission will be the return that the Companies that will be reflected in rates over the rate period, until such time as another rate proceeding is filed. It is important to reflect investors' expectations of returns over that time period. Therefore, I conclude that it is appropriate to consider consensus estimates of investors' expectation of yields on Treasury bonds in the CAPM analysis.

Q. Does Staff misunderstand your testimony with respect to the use of 30-year Treasury bonds in the CAPM?

A. Yes. Staff has misquoted my testimony on this issue in this case and in several prior cases, including Case 16-G-0369 and Case 16-G-0257. Staff suggests that in my Direct Testimony I conclude that "all utility equity investors have an investment horizon of 30 years."[66] Staff then states that this conclusion is unsubstantiated and erroneous. Staff provides no citation for where this conclusion appears in my Direct Testimony.

Q. What is your testimony with respect to the use of the yield on 30-year Treasury bonds as the risk-free rate?

A. It has been my testimony in each of the cases referenced above that it is important to select the term (or maturity) that best matches the life of the underlying investment.[67] I also cite to Morningstar, which notes that the Treasury security relied on should match the time horizon of what is being valued, and that the time horizon is a function of the investment, not the investor. In each case where Staff has asserted this conclusion from my testimony, I have provided this same clarification in response to Staff's assertion. My testimony does not suggest that equity investors have an

[66] Prepared Testimony of the Staff Finance Panel, at 115. *See* also Case 16-G-0369 Prepared Testimony of Staff Finance Panel, at 95, and Case 16-G-0257 Prepared Testimony of Staff Finance Panel, at 91.

[67] Direct Testimony of Ann E. Bulkley, at 72.

investment horizon of 30 years, nor does my testimony suggest that this would be a relevant factor in determining the appropriate security for the risk-free rate even if investors' time horizons were that long.

Q. Please summarize Staff's concern with your use of Bloomberg's Beta coefficients in the CAPM analysis.

A. Staff contends that the Commission has consistently preferred Value Line's Beta coefficients because they are more accurate, and they reflect the exercise of expert judgment. To support this position, Staff cites two articles analyzing the relative accuracy of Value Line's Beta coefficients compared to other sources.[68]

Q. Why is it reasonable to rely on Bloomberg's Beta coefficients?

A. It is reasonable to consider several measures of market conditions in estimating the ROE. As noted in my Direct Testimony, the Bloomberg Beta coefficient is widely used, and because it is based on a ten-year period as compared to Value Line's five-year period, it mitigates the exogenous effect of the TCJA on utility betas.[69] Neither of the academic studies cited by Staff were published after the Senate passed the TCJA on December 2, 2017. As illustrated by Figure 9 in my Direct Testimony, the performance

[68] Prepared Testimony of the Staff Finance Panel, at 111-112.

[69] Direct Testimony of Ann E. Bulkley, at 74-75.

of utility stocks deviated substantially from the performance of the broader market following the passage of the TCJA.

As Staff explained, Value Line's Beta coefficients are calculated over a five-year period.[70] As shown in Figure 4 below, in the last five years, the performance of the S&P Utilities Index has been significantly less correlated with the performance of the S&P 500 than the long-term historical correlation of the two indices. However, the relative volatility of the two indices is essentially unchanged. This indicates that the relative riskiness of utility stocks is essentially unchanged following the passage of the TCJA, even though utility stocks' Beta coefficients have declined. Thus, CAPM results derived solely from Value Line's Beta coefficients are likely to be understated because they are calculated using the last five years of financial data. In addition, this is inconsistent with Staff's DCF analysis, which suggests there has been no change in the cost of equity since the Companies' last case.[71]

[70] Prepared Testimony of the Staff Finance Panel, at 56.

[71] In this case, Staff's DCF result is 8.22 percent. In 16-G-0058 & 16-G-0059 Staff's DCF result was also 8.22 percent.

Figure 4: Beta Input Analysis

	Correlation (S&P Utilities Index Compared to S&P 500)	Relative Volatility (S&P Utilities Index Compared to S&P 500)
Current Five-Year Value (June 2014 – June 2019)	0.34	1.0304
Long-Term Historical Value (2000 – June 2019)	0.56	1.0318

Q. What effect does the decline in Beta coefficients have on the results of Staff's CAPM analysis?

A. As presented in Figure 5 below, the result of Staff's CAPM analysis in the current case are approximately 120 basis points lower than the result of Staff's CAPM analysis in the Companies' previous rate cases. The nearly 20.00 percent decline in Beta coefficients between those two rate cases is responsible for the vast majority of the decline in the results of Staff's CAPM analysis. The slight decrease in Staff's projected market return is mostly offset by a slight increase in Staff's calculated risk-free rate.

Figure 5: Staff Results Comparison

	Risk-Free Rate	**Beta**	**Market Return**	**CAPM Result**
16-G-0058 & 16-G-0059	2.32%	0.75	11.17%	9.24%
19-G-0309 & 19-G-0310	2.55%	0.61	10.88%	8.04%
Difference	0.23%	-0.14	-0.29%	
Impact on CAPM Result	0.05% Increase	1.08% Decrease	0.22% Decrease	1.20% Decrease

Q. Has Value Line recognized the importance of applying its Beta coefficients in the context of other information?

A. Yes, in Value Line's guidance it states "We propose Beta should be used as one factor in the equity analysis framework. Investors should also look at our Safety rank and Price Stability score when making investment decisions."[72] As shown in Exhibit __ (AEB-4R), since the Companies' last rate case, the proxy companies' Value Line Beta coefficients have decreased by more than twice as much as the proxy companies' Safety rank and Price Stability scores, suggesting that the Beta coefficients may be inconsistent with other risk measures. Consistent with Value Line's guidance, this check of reasonableness suggests other inputs should be considered. As such, I have also considered Bloomberg Beta coefficients in my analysis.

Q. What research did Staff cite regarding Value Line's Beta coefficients?

A. Staff cites a 2011 finance text by H. Kent Baker and Gerald S. Martin titled 'Capital Structure and Corporate Financing Decisions: Theory, Evidence, and Practice" and an 1983 article by Diana R. Harrington entitled "Whose Beta Is Best?" to suggest that the Value Line estimate of Beta coefficients is superior.[73]

[72] Value Line "Using Beta," Cueter, Andrew J., October 2, 2012, http://www.valueline.com/Tools/Educational_Articles/Stocks/Using_Beta.aspx.

[73] Prepared Testimony of the Staff Finance Panel, at 111-113.

Q. Do those sources demonstrate that Staff's Beta coefficients are superior to those reflected in your CAPM analysis?

A. No, they do not. As an initial matter, I've included Value Line's Beta coefficients in my CAPM analysis. As such, I agree that Value Line Beta coefficients provide relevant information to investors. However, as with all elements of the ROE determination, I believe it is incumbent on the practitioner to review the results of the analyses and exercise judgment as to how to weight those results in the overall ROE determination. As I demonstrated in Figure 9 of my Direct Testimony, the performance of utility stocks deviated substantially from the performance of the broader market following the passage of the TCJA and therefore I also considered Bloomberg Beta coefficients to mitigate the exogenous effect of the TCJA on utility betas. Neither of the academic studies cited by Staff were published after the Senate passed the TCJA on December 2, 2017. In fact, Diana R. Harrington article, "Whose Beta Is Best?" was published only seven months after the introduction of the Bloomberg terminal and therefore does not consider Bloomberg estimates of Beta coefficients.[74]

Further, the conclusions of the articles are consistent with the process I have used in my application of Beta coefficients in my CAPM analysis. Baker and Martin criticize estimates that "involve no input from analysts" and go

[74] McCracken, Harry, "How the Bloomberg Terminal Made History–And Stays Ever Relevant," Fast Company, October 6, 2015, https://www.fastcompany.com/3051883/the-bloomberg-terminal.

on to conclude "Value Line's adjustments also appear informed by more than the type of simple smoothing done by Bloomberg, which uses the following formula: adjusted beta = historic beta (0.67) +0.33" [75]. However, the smoothing performed by Value Line is similar (adjusted beta = historic beta (0.67) +0.35). And in my application of Bloomberg Beta coefficients, I adjusted the measurement period to mitigate the effects of the TCJA as described above. This is consistent with Baker and Martin's conclusion on Beta, "The objective is to capture investors' view of future risks." As such, I disagree with Staff's conclusion that these sources demonstrate that Value Line Beta coefficients are superior to my application of Bloomberg coefficients that were specified with parameters based on an analysis of the market conditions and my informed judgment.

Q. Please summarize Staff's concern with your estimation of the Market Risk Premium.

A. Staff testifies that the use of the Constant Growth DCF model to develop the market return on the S&P 500 is not appropriate because the companies in the S&P 500 cannot sustain their three-to-five year growth rates in perpetuity.[76]

[75] Exhibit ___ (SFP-33) at 3.

[76] Prepared Testimony of the Staff Finance Panel, at 108-109.

Q. What is your response?

A. The calculation of the MRP is based on the return on the broader market, as measured by S&P 500, less the return on a risk-free instrument. The S&P 500 is an index that includes the largest 500 companies by market capitalization. Over time, the specific companies that are included in the S&P 500 Index will vary, but investor expectations of growth and return overall for the index as a whole may not, based on the selection process involved in the index. Therefore, it is reasonable to assume that the average growth of the index could be sustainable in the long run.

Q. Is there additional support for the reasonableness of the market return you have used to calculate the forward-looking market risk premium?

A. Yes, other alternative sources provide reputable forecasts of market returns that are significantly higher than the estimated market return relied on by Staff. In Figure 6, I provide the return on the S&P 500 calculated using earnings growth projections from Bloomberg Professional, Yahoo! Finance, and Standards and Poor's. The calculated returns for the S&P 500 range from 12.23 percent (Bloomberg Professional) to 14.42 percent (Standard and Poor's). By contrast, the historical market return that Staff relied on is 10.88 percent, which is more than 100 basis points below the market return estimates provided by these reputable sources.

Figure 6: S&P 500 Return Estimates[77]

Source	Estimate Date	Dividend Yield	Growth Estimate	S&P 500 Return
Bloomberg Professional	February 28, 2019	1.97%	10.16%	12.23%
Yahoo! Finance	February 28, 2019	1.97%	11.00%	13.08%
Standard and Poor's	February 28, 2019	1.97%	12.33%	14.42%

Q. What are your conclusions regarding the CAPM analysis?

A. In principle, Staff and I agree that the development of the inputs and assumptions for the CAPM should be done on a forward-looking basis, and that *ex-post* analysis can be flawed in some circumstances. Furthermore, both our CAPM analyses contain some historical data where we believe it to be the best estimate in current market conditions. In particular, I note that recent market conditions should not be considered a proxy for future ROE results. As such, I have developed the assumptions used in the CAPM to be forward-looking to the period that KEDNY's and KEDLI's rates will be in effect. In the instance of Beta, I relied on Bloomberg Beta coefficients in addition to Value Line's Beta coefficients in order to account for the effects of the TCJA on utility Beta coefficients.

[77] Bloomberg and Yahoo! Finance do not report a dividend yield for the S&P 500; therefore, the most recent 12-month average dividend yield as of February 2019 reported in the February 28, 2019 S&P 500 Earnings and Estimate Report was used to calculate the total return.

Based on the specification of the methodology, and the support for this methodology that has been offered by other regulatory agencies as a result of recent market conditions, I believe that it is appropriate to give this form of the CAPM an equal weighting in the estimation of the ROE.

E. CREDIT METRICS

Q. Please describe Staff's analysis regarding the effect its recommended common equity ratio and return on equity would have on the Companies' financial integrity.

A. To assess the effect of its recommended capital structure and return on equity would have on the Companies' financial integrity, Staff analyzed a variety of credit metrics that Moody's and S&P utilize when determining the Companies' credit ratings. The results of that analysis are summarized in Exhibit ___ (SFP-10).

Q. Do you have any concerns with Staff's analysis?

A. Yes, I have two primary concerns with Staff's analysis. First, if Staff's recommended return on equity were adopted, it is likely to undermine investors' perceptions of the supportiveness of the regulatory environment in New York. Second, Moody's has explicitly highlighted concerns around the Companies' Cash Flow from Operation Pre-Working Capital to Debt ("CFO pre-WC/Debt") as a factor that could lead to a downgrade. For KEDNY, Moody's projections for CFO pre-WC/Debt are between 11.2

percent and 13.9 percent.[78] While Moody's projections for KEDLI are somewhat higher,[79] Staff's analysis indicates that an 8.20 percent return on equity would produce a CFO pre-WC/Debt metric of 13.6 percent for KEDLI. Both of these factors would increase the chances of a credit rating downgrade, as explained in more detail below.

Q. Please elaborate on your first concern.

A. Moody's rates the Companies' in accordance with its Regulated Electric and Gas Utilities rating methodology, which consists of four primary factors. The first two primary factors, "Regulatory Framework" and "Ability to Recover Costs and Earn Returns," account for 50.00 percent of Moody's overall rating for the Companies. One of the two sub-factors listed under "Regulatory Framework" is "Consistency and Predictability of Regulation". One of the two sub-factors listed under "Ability to Recover Costs and Earn Returns" is "Sufficiency of Rates and Returns".

Q. How would Staff's recommended return on equity affect those factors?

A. Staff's recommended return on equity of 8.20 percent is likely to be viewed by the credit rating agencies as inconsistent, unpredictable, and insufficient because it is 143 basis points below the year-to-date average authorized return on equity for gas distribution utilities across the U.S., and 50 basis

[78] Exhibit ___ (SFP-5), at 2.
[79] Exhibit ___ (SFP-6), at 2.

points lower than any authorized return on equity since at least 1980. Further, as reflected on Exhibit ___ (AEB-7), Staff's recommended ROE is 80 basis points below the lowest authorized ROE of any operating subsidiary of the utilities in my Combined Utility Proxy Group. An authorized ROE of 8.20 percent would be unprecedented, and likely worsen investors' perception of the supportiveness of the Companies' regulatory environment. The Moody's reports for both KEDNY and KEDLI list "Decrease in the New York Public Services Commission's (NYPSC) overall supportiveness" as a factor that could lead to a credit rating downgrade.[80]

Q. Has Staff offered testimony that corroborates your first concern?

A. Yes, Staff has. Staff states that major credit rating agencies "view unpredictability as an additional risk."[81] For context, Staff offered that testimony while criticizing my choice to use projected earnings growth rates in the DCF model. However, it is difficult to understand how the credit rating agencies would view an 80 basis point reduction in the Company's authorized ROE as "predictable." It is highly unlikely that the investment community would be concerned by the use of First Call and Zacks in the DCF model, but not by an 80 basis point decline in the Companies' authorized ROE.

[80] Exhibit ___ (SFP-5), at 2, and Exhibit ___ (SFP-6), at 2.

[81] Prepared Testimony of the Staff Finance Panel, at 92-93.

Q. Please elaborate on your second concern regarding the CFO pre-WC/Debt metric.

A. Staff states that the CFO pre-WC/Debt metric is the "core coverage ratio used by Moody's to assess the financial health of regulated electric and gas utility companies."[82] According to Staff's analysis, an ROE of 8.20 percent would result in a CFO pre-WC/Debt metric of 13.6 percent for KEDLI.[83]

Q. Has Moody's corroborated Staff's assessment of the importance of the CFO pre-WC/Debt metric?

A. Yes. Moody's has explained that a CFO pre-WC/Debt metric that no longer trends toward the mid-teens in the near-term, and to the high teens over the longer term is a factor that could lead to a credit rating downgrade for KEDNY.[84] Given Moody's projections of a CFO pre-WC/Debt metric between 11.2 percent and 13.9 percent, it is difficult to reconcile how an ROE of 8.20 percent will help KEDNY raise its CFO pre-WC/Debt metric to range sufficient to avoid a downgrade. In addition, Moody's indicated that its "Stable" outlook for KEDLI reflects its expectation that "following the approval of a new rate plan, KEDLI will be able to achieve CFO pre-WC/debt trending towards the high-teens in percentage terms." Moody's

[82] Prepared Testimony of the Staff Finance Panel, at 43.
[83] Exhibit ___ (SFP-6), at 2.
[84] Exhibit ___ (SFP-5), at 2.

further explained that a CFO pre-WC/Debt metric consistently below the high teens is a factor that could lead to a credit rating downgrade for KEDLI.[85]

Q. In light of Staff's assessment that an ROE of 8.20 percent would produce a CFO pre-WC/Debt metric below the high teens, how would Staff's recommended ROE impact KEDLI's credit rating?

A. As described in my Direct Testimony, the TCJA has already put downward pressure on the credit metrics of many utilities and their holding companies.[86] The Companies' are no exception. In fact, earlier this year, Moody's downgraded the senior unsecured rating and issuer ratings of KEDNY and KEDLI, citing weakness in key credit metrics. If it were adopted, Staff's recommended ROE of 8.20 percent, would further erode the Companies' performance on a core Moody's metric, and would likely increase the prospects of another credit rating downgrade.

Q. What are your conclusions regarding the impact Staff's recommended return on equity would have on the Companies' financial integrity?

A. Staff's recommended return on equity is likely to have a negative effect on the Company's financial integrity. It is unpredictable, inconsistent, and insufficient, which would materially worsen investors' perceptions of the

[85] Exhibit ___ (SFP-10), at 3.
[86] Direct Testimony of Ann E. Bulkley, at 31-38.

credit supportiveness of the Companies' regulatory environment. Further, due to the specific factors that Moody's considers when determining the Companies' credit rating, it would particularly strain the credit rating of the Companies.

F. REGULATORY FRAMEWORK AND BUSINESS RISKS

Q. Please summarize Staff's position regarding the Companies' business risks.

A. Staff relies on credit rating agencies' commentary on the risk of the utility segment as a whole to support conclusions regarding the risk of KEDNY and KEDLI. Staff's review of the Companies' business risk is as compared to the overall industry risk for electric utilities and does not provide any evidence as to the relative risk of the Companies and the proxy group.[87] Because the ROE is estimated using the proxy group results as a surrogate for the investor required return, it is important that any analysis of the credit supportiveness and business risk of the Companies be conducted relative to the proxy group.

Q. Is Staff's business risk analysis consistent with the development of its proxy group?

A. No, it is not. Staff supports the use of an electric proxy group on the basis that the parent company of KEDNY and KEDLI has significant electric

[87] *Id*., at 123-133.

operations, and therefore the use of an electric utility proxy group is appropriate, even though KEDNY and KEDLI are natural gas distribution companies. However, in the discussion of business risk, Staff concludes, without an analysis of their specific proxy group, that KEDNY and KEDLI are less risky than most electric utility companies.[88] Based on these differences in risk, it would seem more appropriate in setting the ROE for the Companies to consider the returns of a proxy group that includes natural gas distribution companies that have a more similar risk profile to KEDNY and KEDLI.

Q. Have you conducted any analysis of the risk of your proxy group as compared with KEDNY and KEDLI?

A. Yes. As discussed in my Direct Testimony, I examined two rankings performed by the investment community of U.S. regulatory commissions (one by Regulatory Research Associates, or "RRA", and one by Standard & Poor's, or "S&P"). As noted in my Direct Testimony, RRA accords New York an "Average/1" rating, which is only the fourth best ranking (out of nine).[89] S&P ranks New York in the exact middle of the scale in terms of credit supportiveness, in contrast to the proxy group's overall mean ranking which is slightly above average. This suggests that, while New York may

[88] *Id.,* at 16.

[89] Regulatory Research Associates, https://www.snl.com/web/client?auth=inherit#industry/CommissionProfiles (September 9, 2019).

have a variety of ratemaking policies that lower the risks faced by the Companies, the Companies' regulatory environment is below average when compared to the proxy group. Further, investors' perception of the Companies' regulatory environment is likely to worsen if the Commission authorizes Staff's recommended ROE of 8.20 percent for the Companies', which is 48 basis points lower than the lowest ROE authorized by the Commission in the past 35 years.

Q. Staff notes that revenue decoupling mechanisms reduce revenue risk. Have you considered the effect of the Companies' revenue decoupling mechanisms on the required ROE?

A. Yes, I have. The ROE recommendation is established for a company based on its risk relative to the proxy group. Therefore it is necessary to review the similarity of the Companies' revenue decoupling mechanisms to the alternative rate mechanisms that have been implemented by the proxy companies. Exhibit__(AEB-7) summarizes the alternative rate mechanisms that have been implemented by the proxy companies. As shown in Exhibit __ (AEB-7), and as described in my Direct Testimony, approximately 60.00 percent of the jurisdictions where the proxy companies operate have approved some form of adjustment mechanism (*i.e.*, formula rate plan, revenue decoupling mechanism, straight fixed-variable rate design) to reduce the risk of revenue shortfalls. In addition, the authorized ROEs for the proxy group range from 9.00 percent to 10.50 percent. Therefore, it is

difficult to reconcile Staff's recommendation, which is 80 basis points below the *lowest* authorized return for any of the operating companies of the proxy companies, given the returns available to substantially comparable companies.

Q. Do you agree with Staff's suggestion that the Companies' deferral mechanisms should result in a lower ROE?

A. No, I do not. Since the ROE is established based on the expected returns for a proxy group of comparable companies, the analysis of business risk is necessarily a comparative analysis. The relevant question is not how much of the Company's risk has been reduced, but rather how much risk do the Companies have relative to the proxy group. In that regard, Staff has provided no record evidence. While Staff suggests that most of the Company's revenues and costs are subject to true-ups and deferrals, the recovery of gas and fuel costs are a pass-through cost for nearly all utility companies. Fuel costs are the single largest cost for most utilities. Since nearly all utilities are allowed to pass through purchased power and gas costs directly to customers, there is no risk differential between the Companies and the proxy companies on the recovery of this cost. As noted previously, Staff has not provided any record evidence as to the level of recovery or true-up of the Company's remaining operating costs as compared with the peer group to assess the Company's relative risk profile. As described in the Company's Revenue Requirements Panel Rebuttal

Testimony, Staff has proposed a series of downward-only true-ups for costs associated with full-time equivalents, Service Company IT rent expense, paving, GBE, and credit cards. To the extent any of Staff's proposed downward-only true-ups are implemented, the Companies' will remain exposed to risks associated with any cost overruns or inflation above a certain threshold. Conversely, Staff proposes to eliminate the Company's property tax reconciliation and change KEDNY's city-state construction reconciliation from a 90/10 sharing to an 80/20 sharing. The effect of such asymmetric cost trackers provides little, if any, risk mitigation.

Q. Have the regulatory mechanisms available in New York allowed the Companies to earn their authorized return on equity historically?

A. No. As shown on Exhibit ___ (SFP-5), at 5, KEDNY has under-earned relative to its authorized return on equity in four of the last five years. As shown on Exhibit ___ (SFP-6), at 5, KEDLI has under-earned relative to its authorized return on equity in three of the last five years. While both Companies' achieved returns have recovered in 2018, this sustained historical under-earning casts doubt on Staff's claim that the suite of regulatory mechanisms available in New York insulate the Companies from financial risk.

Q. Did Staff acknowledge the effect of the Companies' moratorium on processing new natural gas service applicants on investors' return requirements?

A. Yes, Staff did. Specifically, Staff stated that this development could "modestly and negatively impact investors return requirements" because it will limit the Companies' future growth opportunities.[90] While Staff argues that the impact of this development may be moderated by adjustments to the design of the Companies' Revenue Decoupling Mechanism, Staff did acknowledge that this is a business risk faced by the Companies.

Q. Did Staff discuss the additional risk faced by the Companies if a multi-year rate plan is approved by the Commission?

A. Yes. Staff appears to agree that investors should be compensated with an ROE premium for the additional risks associated with "staying out" of the rate case process. Specifically, Staff stated:

[90] Prepared Testimony of the Staff Finance Panel, at 75.

> It is important to note, however, that when the Commission evaluates the reasonableness of a multi-year rate plan's ROE, it recognizes the necessity of a premium above the ROE produced using Staff's methodology in recognition of the increased financial and business risks that are inherent when a company is operating within the parameters of a multi-year rate plan. By agreeing not to file for new rates for three years, the Companies reduced their ability to adjust rates in response to changing conditions. Accordingly, locking in rates for an extended period somewhat diminished the Companies' flexibility and thereby increased their level of risk. Additionally, the forecasts of expense items in years two and three of a multi-year rate plan are inherently more inaccurate and thus expose a company to additional operating risk.[91]

While Staff has not quantified the ROE premium that would adequately compensate investors for those incremental risks, I determined that a 50-basis point increase to the Companies' authorized ROE would be appropriate if a multi-year rate plan is approved by the Commission.

Q. What are your conclusions regarding the Companies' business risk relative to the proxy group?

A. As discussed in my Direct Testimony, New York's relative regulatory rankings from a credit perspective, coupled with the heightened level of business risk faced by the Companies due to their capital spending plans, and Moody's concerns about weakened credit metrics demonstrate that the Companies' business risk is above average compared to the proxy

[91] *Id*., at 73-74.

companies.[92] In addition, Staff recognizes that there is an incremental risk due to the effect of the Companies' moratorium on processing new natural gas service.[93] That elevated risk profile increases the importance of setting a return for the Companies that is within the range of reasonableness as established by the returns for the two proxy groups that I used in formulating my return recommendation.[94]

IV. CONCLUSIONS AND RECOMMENDATION

Q. What is your conclusion regarding a fair return on equity for KEDNY and KEDLI?

A. I continue to support a recommended return on equity for the Companies' within the range from 9.65 percent and 10.55 percent based on the analyses my Direct Testimony and the updates proposed to Staff's analyses that are presented in Figure 7, below.

[92] Direct Testimony of Ann E. Bulkley, at 91-94.
[93] Prepared Testimony of the Staff Finance Panel, at 75.
[94] Direct Testimony of Ann E. Bulkley, at 82.

Figure 7: Summary of Staff's Adjusted Results[95]

	Adjustment	Mean DCF Result	Mean Staff CAPM[96]	ROE Result	Incremental Change (bps)
	Staff Estimate	8.22%	8.04%	8.16%	
1	Add gas utilities to Staff Proxy Group	8.47%	8.11%	8.35%	+19 bps
2	Use Value Line and Bloomberg Beta Coefficients	8.47%	8.33%	8.42%	+7 bps
3	Use the more precise XIRR function	8.60%	8.33%	8.51%	+9 bps
4	Apply Market Risk Premium Estimate based on S&P 500 market return	8.60%	10.43%	9.21%	+70 bps
5	Apply GDP Growth Rate for Long-Term Growth in the DCF Model	8.83%	10.43%	9.36%	+15 bps
6	Evenly Weight the DCF/CAPM Analyses	8.83%	10.43%	9.63%	+27 bps
Total Change				9.63%	+147 bps

The Companies' proposed ROE of 9.65 percent is supported by the results of the DCF and CAPM methods, and the specific risks to which the Companies are exposed. In contrast, Staff's recommendation of 8.20 percent is lower than any return that has been authorized by any regulatory

[95] *See* Exhibit ___ (AEB-1R).

[96] CAPM relies on the mean Beta to be consistent with Staff's use of the mean DCF result.

commission and is significantly below the returns authorized by this Commission in recent cases. Therefore, Staff's recommendation does not result in a return that is commensurate with the return on other companies of similar risk. Furthermore, Staff's recommendation is likely to be viewed by rating agencies as unsupportive and could result in a further downgrade of Moody's credit rating for KEDLI, in particular. As a result, Staff's recommended ROE should be rejected.

Q. Does this conclude your Rebuttal Testimony?

A. Yes, it does.

INDEX OF EXHIBITS

Exhibit __ (AEB-1R):	Summary of Staff Adjusted Results / Discounted Cash Flow Analysis
Exhibit __ (AEB-2R):	Proxy Group Analysis / Staff Finance Panel's Proxy Group / Credit Rating and Business Segment Data
Exhibit __ (AEB-3R):	Response to IR DPS-894
Exhibit __ (AEB-4R):	Proxy Group Analysis / Staff Finance Panel's Proxy Group / Value Line Risk Measures

Exhibit ___ (AEB-1R)

Summary of Staff Adjusted Results
Discounted Cash Flow Analysis

Summary of Staff's Adjusted Results

Description	[1] Unadjusted Staff Results	[2] Step One: Add Gas Utilities to Proxy Group	[3] Step Two: Add 10-Year Bloomberg Betas	[4] Step Three: Apply More Precise XIRR Function	[5] Step Four: Apply Bulkley MRP Estimate	[6] Step Five Evenly Weight DCF / CAPM Results	[7] Step Six: Bulkley GDP Growth Rate in DCF	[8] Companies' ROE Request
Proxy Group DCF ROE	8.22%	8.47%	8.47%	8.60%	8.60%	8.60%	8.83%	
Market Return Estimate	10.88%	10.88%	10.88%	10.88%	13.90%	13.90%	13.90%	
Risk-Free Rate	2.55%	2.55%	2.55%	2.55%	2.55%	2.55%	2.55%	
Market Risk Premium	8.33%	8.33%	8.33%	8.33%	11.35%	11.35%	11.35%	
Beta	0.61	0.62	0.65	0.65	0.65	0.65	0.65	
Traditional CAPM ROE	7.63%	7.72%	7.97%	7.97%	9.93%	9.93%	9.93%	
Zero Beta CAPM ROE	8.45%	8.51%	8.70%	8.70%	10.92%	10.92%	10.92%	
Generic (Average) CAPM ROE	8.04%	8.11%	8.33%	8.33%	10.43%	10.43%	10.43%	
Weighted ROE Result	**8.16%**	**8.35%**	**8.42%**	**8.51%**	**9.21%**	**9.51%**	**9.63%**	**9.65%**
Incremental Change		**0.19%**	**0.07%**	**0.09%**	**0.70%**	**0.30%**	**0.12%**	

Notes:
[1] Source: Exhibit ___ (SFP-17), at 2; Exhibit ___ (SFP-20), at 1
[2] Source: Exhibit ___ (AEB-1R), at 2-3
[3] Source: Exhibit ___ (AEB-1R), at 2-3
[4] Source: Exhibit ___ (AEB-1R), at 2-3
[5] Source: Bloomberg Professional, methodology from Exhibit ___ (AEB-5) updated consistent with timing of Staff Market Return analysis
[6] Source: Equals column [5]
[7] Source: Exhibit ___ (AEB-1R), at 4; Exhibit ___ (AEB-2)
[8] Source: Direct Testimony of Ann E. Bulkley, at 102

Discounted Cash Flow Analysis
Staff Finance Panel's Proxy Group + Gas Utilities
As of June 30, 2019

Company	Ticker	Value Line Beta	Bberg Beta	Price	EPS 2022-24	DPS 2019	DPS 2020	DPS 2022-24	BPS 2019	BPS 2020	BPS 2022-24	# of Shares 2019	# of Shares 2022-24	DPS Growth 2022-24
Allete Inc.	ALE	0.65	0.71	81.97	4.25	2.35	2.46	2.85	43.05	44.35	48.50	51.75	51.75	5.03%
Alliant Energy Corp	LNT	0.60	0.71	47.54	2.80	1.42	1.50	1.74	21.80	23.05	27.55	240.00	250.00	5.07%
Ameren Corp.	AEE	0.60	0.66	73.69	4.25	1.93	2.06	2.55	32.95	35.10	40.75	246.50	255.00	7.37%
American Electric Power Co. Inc.	AEP	0.55	0.64	86.12	5.00	2.72	2.88	3.40	40.05	41.65	47.75	494.65	518.00	5.69%
Avista Corp	AVA	0.65	0.71	42.61	2.50	1.55	1.61	1.80	28.55	29.25	32.00	67.00	71.00	3.79%
Black Hills Corp	BKH	0.80	0.77	75.34	4.50	2.05	2.17	2.60	38.10	39.90	45.00	60.50	61.00	6.21%
CMS Energy Corp	CMS	0.55	0.66	55.94	3.25	1.53	1.64	2.00	17.90	19.35	24.50	285.00	297.00	6.84%
Consolidated Edison Inc.	ED	0.45	0.54	86.36	5.08	2.96	3.06	3.40	53.00	54.50	59.75	334.00	338.00	3.57%
Dominion Energy Inc.	D	0.55	0.62	76.04	5.00	3.67	3.76	4.05	34.75	35.90	39.50	808.00	840.00	2.51%
Duke Energy Corp.	DUK	0.50	0.55	88.29	5.75	3.75	3.83	4.15	61.65	63.20	68.50	733.00	755.00	2.71%
Edison International	EIX	0.60	0.68	62.55	5.50	2.45	2.45	2.70	33.60	35.15	41.25	325.81	325.81	3.29%
Entergy Corporation	ETR	0.60	0.67	97.22	6.25	3.66	3.74	4.00	51.35	53.15	60.25	200.00	210.00	2.27%
Eversource Energy	ES	0.60	0.67	72.85	4.25	2.14	2.26	2.65	37.70	39.50	46.75	318.50	350.00	5.45%
First Energy Corp	FE	0.65	0.69	41.68	3.25	1.52	1.60	1.90	14.90	16.10	20.00	540.00	550.00	5.90%
Hawaiian Electric Industries	HE	0.60	0.66	41.8	2.50	1.28	1.32	1.50	20.55	21.45	24.50	109.00	113.00	4.35%
IDACORP Inc.	IDA	0.60	0.74	100.61	5.25	2.56	2.72	3.20	48.80	50.60	56.50	50.40	50.40	5.57%
MGE Energy, Inc.	MGEE	0.55	0.68	68.72	3.80	1.38	1.45	1.65	24.55	26.00	31.95	35.00	36.00	4.40%
NextEra Energy, Inc.	NEE	0.60	0.65	196.56	11.25	5.00	5.65	7.00	69.05	72.50	84.50	535.00	535.00	7.40%
Northwestern Corp	NWE	0.60	0.70	70.76	4.00	2.30	2.40	2.70	39.70	40.80	44.00	50.50	51.10	4.00%
OGE Energy Corp.	OGE	0.80	0.76	42.31	2.75	1.54	1.65	1.95	20.55	21.15	23.25	200.00	200.00	5.73%
Pinnacle West Capital Corp.	PNW	0.55	0.68	95.42	5.75	3.04	3.22	3.80	48.30	50.00	55.50	112.50	114.50	5.68%
PNM Resources Inc.	PNM	0.65	0.78	47.47	2.75	1.18	1.24	1.50	22.05	23.30	27.50	79.65	84.00	6.55%
Portland General Electric Co.	POR	0.60	0.68	52.77	3.00	1.52	1.62	1.95	29.00	29.90	32.75	89.40	90.00	6.38%
PPL Corp	PPL	0.70	0.63	30.88	2.75	1.65	1.66	1.80	17.35	18.30	21.75	770.00	780.00	2.74%
Sempra Energy	SRE	0.75	0.71	131.39	9.00	3.87	4.20	5.25	60.00	64.80	77.00	290.00	320.00	7.72%
Southern Co.	SO	0.50	0.54	53.47	3.75	2.46	2.54	2.78	26.00	26.90	30.00	1045.00	1085.00	3.06%
WEC Energy Group	WEC	0.50	0.59	79.95	4.50	2.36	2.50	3.00	32.05	33.40	36.75	315.50	315.50	6.27%
Xcel Energy Inc.	XEL	0.50	0.59	57.4	3.25	1.62	1.72	2.05	24.85	25.95	29.50	515.50	521.50	6.03%
Atmos Energy Corporation	ATO	0.65	0.69	102.30	5.60	2.10	2.24	2.70	48.30	49.90	56.05	120.00	145.00	6.42%
New Jersey Resources Corporation	NJR	0.70	0.73	49.06	2.50	1.17	1.21	1.33	17.25	18.30	21.85	88.00	89.00	3.20%
Northwest Natural Gas Company	NWN	0.60	0.69	67.33	3.50	1.93	1.97	2.20	26.55	28.20	29.40	30.00	32.00	3.75%
ONE Gas, Inc.	OGS	0.65	0.64	88.19	4.75	2.00	2.16	2.65	41.05	42.75	47.90	53.00	55.00	7.05%
South Jersey Industries, inc.	SJI	0.80	0.76	32.24	2.40	1.20	1.25	1.40	16.50	17.20	20.00	94.00	100.00	3.85%
Southwest Gas Corporation	SWX	0.70	0.75	84.92	5.85	2.18	2.30	2.60	45.35	48.20	59.50	54.00	58.00	4.17%
Spire, Inc.	SR	0.65	0.64	83.82	5.00	2.37	2.46	2.67	49.00	50.00	52.75	51.00	55.00	2.77%
Average		**0.62**	**0.67**											**4.94%**
Average (Excl. Gas Utilities)		**0.61**	**0.67**											**5.06%**
Average (Gas Utilities Only)		**0.68**	**0.70**											**4.46%**

Note: Replicated analysis based on Staff Finance Panel Filed Exhibits, as filed on August 30, 2019

Discounted Cash Flow Analysis
Staff Finance Panel's Proxy Group + Gas Utilities
As of June 30, 2019

Company	Ticker	Retention Rate 2023	Return on Equity 2023	B x R	Increase in Shares	MBR 2019	MBR-1 V Factor	Accrection/ Dilution Growth S x V	Sustainable Growth	IRR-Based Long-Form ROE	XIRR-Based Long-Form ROE
Allete Inc.	ALE	32.94%	8.89%	2.93%	0.00%	1.90	0.90	0.00%	2.93%	6.05%	6.14%
Alliant Energy Corp	LNT	37.86%	10.47%	3.96%	1.03%	2.18	1.18	1.21%	5.17%	8.23%	8.35%
Ameren Corp.	AEE	40.00%	10.69%	4.28%	0.85%	2.24	1.24	1.05%	5.33%	8.19%	8.30%
American Electric Power Co. Inc.	AEP	32.00%	10.71%	3.43%	1.16%	2.15	1.15	1.33%	4.76%	8.10%	8.23%
Avista Corp	AVA	28.00%	7.93%	2.22%	1.46%	1.49	0.49	0.72%	2.94%	6.75%	6.87%
Black Hills Corp	BKH	42.22%	10.20%	4.31%	0.21%	1.98	0.98	0.20%	4.51%	7.45%	7.55%
CMS Energy Corp	CMS	38.46%	13.79%	5.30%	1.04%	3.13	2.13	2.20%	7.51%	10.27%	10.41%
Consolidated Edison Inc.	ED	33.07%	8.63%	2.85%	0.30%	1.63	0.63	0.19%	3.04%	6.58%	6.69%
Dominion Energy Inc.	D	19.00%	12.86%	2.44%	0.98%	2.19	1.19	1.16%	3.60%	8.33%	8.52%
Duke Energy Corp.	DUK	27.83%	8.51%	2.37%	0.74%	1.43	0.43	0.32%	2.69%	6.97%	7.12%
Edison International	EIX	50.91%	13.69%	6.97%	0.00%	1.86	0.86	0.00%	6.97%	10.41%	10.58%
Entergy Corporation	ETR	36.00%	10.59%	3.81%	1.23%	1.89	0.89	1.10%	4.91%	8.41%	8.55%
Eversource Energy	ES	37.65%	9.35%	3.52%	2.39%	1.93	0.93	2.22%	5.74%	8.73%	8.85%
First Energy Corp	FE	41.54%	16.84%	6.99%	0.46%	2.80	1.80	0.83%	7.82%	11.34%	11.53%
Hawaiian Electric Industries	HE	40.00%	10.43%	4.17%	0.91%	2.03	1.03	0.94%	5.11%	8.12%	8.24%
IDACORP Inc.	IDA	39.05%	9.46%	3.70%	0.00%	2.06	1.06	0.00%	3.70%	6.47%	6.56%
MGE Energy, Inc.	MGEE	56.58%	12.30%	6.96%	0.71%	2.80	1.80	1.27%	8.23%	9.99%	10.08%
NextEra Energy, Inc.	NEE	37.78%	13.65%	5.16%	0.00%	2.85	1.85	0.00%	5.16%	8.12%	8.23%
Northwestern Corp	NWE	32.50%	9.21%	2.99%	0.30%	1.78	0.78	0.23%	3.22%	6.63%	6.74%
OGE Energy Corp.	OGE	29.09%	12.01%	3.50%	0.00%	2.06	1.06	0.00%	3.50%	7.56%	7.70%
Pinnacle West Capital Corp.	PNW	33.91%	10.54%	3.57%	0.44%	1.98	0.98	0.43%	4.01%	7.46%	7.58%
PNM Resources Inc.	PNM	45.45%	10.28%	4.67%	1.34%	2.15	1.15	1.54%	6.21%	8.75%	8.86%
Portland General Electric Co.	POR	35.00%	9.30%	3.25%	0.17%	1.82	0.82	0.14%	3.39%	6.65%	6.76%
PPL Corp	PPL	34.55%	13.01%	4.49%	0.32%	1.78	0.78	0.25%	4.75%	9.74%	9.98%
Sempra Energy	SRE	41.67%	12.02%	5.01%	2.49%	2.19	1.19	2.96%	7.97%	11.02%	11.18%
Southern Co.	SO	25.87%	12.73%	3.29%	0.94%	2.06	1.06	1.00%	4.29%	8.79%	8.99%
WEC Energy Group	WEC	33.33%	12.44%	4.15%	0.00%	2.49	1.49	0.00%	4.15%	7.38%	7.50%
Xcel Energy Inc.	XEL	36.92%	11.25%	4.15%	0.29%	2.31	1.31	0.38%	4.53%	7.57%	7.68%
Atmos Energy Corporation	ATO	51.79%	10.18%	5.27%	4.84%	2.12	1.12	5.42%	10.69%	12.48%	12.59%
New Jersey Resources Corporation	NJR	46.80%	11.78%	5.51%	0.28%	2.84	1.84	0.52%	6.03%	8.22%	8.31%
Northwest Natural Gas Company	NWN	37.14%	11.99%	4.45%	1.63%	2.54	1.54	2.50%	6.95%	9.53%	9.65%
ONE Gas, Inc.	OGS	44.21%	10.10%	4.47%	0.93%	2.15	1.15	1.07%	5.54%	7.99%	8.09%
South Jersey Industries, inc.	SJI	41.67%	12.30%	5.13%	1.56%	1.95	0.95	1.49%	6.61%	10.10%	10.27%
Southwest Gas Corporation	SWX	55.56%	10.18%	5.65%	1.80%	1.87	0.87	1.57%	7.23%	9.61%	9.72%
Spire, Inc.	SR	46.60%	9.56%	4.46%	1.91%	1.71	0.71	1.35%	5.81%	8.42%	8.53%
Average		**38.37%**	**11.08%**	**4.27%**	**0.93%**	**212.42%**	**112.42%**	**1.02%**	**5.29%**	**8.47%**	**8.60%**
Average (Excl. Gas Utilities)		**36.40%**	**11.13%**	**4.09%**	**0.70%**	**211.29%**	**111.29%**	**0.77%**	**4.86%**	**8.22%**	**8.35%**
Average (Gas Utilities Only)		**46.25%**	**10.87%**	**4.99%**	**1.85%**	**216.90%**	**116.90%**	**1.99%**	**6.98%**	**9.48%**	**9.59%**

Note: Replicated analysis based on Staff Finance Panel Filed Exhibits, as filed on August 30, 2019

Discounted Cash Flow Analysis
Staff Finance Panel's Proxy Group + Gas Utilities
Adjusted to Reflect Bulkley's Long-Term Growth Estimate
As of June 30, 2019

Company	Ticker	Price	DPS 2019	DPS 2020	DPS 2022-24	DPS Growth 2022-24	Bulkley GDP Growth Rate	XIRR-Based Long-Form ROE
Allete Inc.	ALE	81.97	2.35	2.46	2.85	5.03%	5.56%	8.54%
Alliant Energy Corp	LNT	47.54	1.42	1.50	1.74	5.07%	5.56%	8.70%
Ameren Corp.	AEE	73.69	1.93	2.06	2.55	7.37%	5.56%	8.51%
American Electric Power Co. Inc.	AEP	86.12	2.72	2.88	3.40	5.69%	5.56%	8.95%
Avista Corp	AVA	42.61	1.55	1.61	1.80	3.79%	5.56%	9.22%
Black Hills Corp	BKH	75.34	2.05	2.17	2.60	6.21%	5.56%	8.51%
CMS Energy Corp	CMS	55.94	1.53	1.64	2.00	6.84%	5.56%	8.62%
Consolidated Edison Inc.	ED	86.36	2.96	3.06	3.40	3.57%	5.56%	8.96%
Dominion Energy Inc.	D	76.04	3.67	3.76	4.05	2.51%	5.56%	10.22%
Duke Energy Corp.	DUK	88.29	3.75	3.83	4.15	2.71%	5.56%	9.66%
Edison International	EIX	62.55	2.45	2.45	2.70	3.29%	5.56%	9.31%
Entergy Corporation	ETR	97.22	3.66	3.74	4.00	2.27%	5.56%	9.13%
Eversource Energy	ES	72.85	2.14	2.26	2.65	5.45%	5.56%	8.68%
First Energy Corp	FE	41.68	1.52	1.60	1.90	5.90%	5.56%	9.49%
Hawaiian Electric Industries	HE	41.8	1.28	1.32	1.50	4.35%	5.56%	8.65%
IDACORP Inc.	IDA	100.61	2.56	2.72	3.20	5.57%	5.56%	8.27%
MGE Energy, Inc.	MGEE	68.72	1.38	1.45	1.65	4.40%	5.56%	7.57%
NextEra Energy, Inc.	NEE	196.56	5.00	5.65	7.00	7.40%	5.56%	8.60%
Northwestern Corp	NWE	70.76	2.30	2.40	2.70	4.00%	5.56%	8.85%
OGE Energy Corp.	OGE	42.31	1.54	1.65	1.95	5.73%	5.56%	9.54%
Pinnacle West Capital Corp.	PNW	95.42	3.04	3.22	3.80	5.68%	5.56%	8.99%
PNM Resources Inc.	PNM	47.47	1.18	1.24	1.50	6.55%	5.56%	8.25%
Portland General Electric Co.	POR	52.77	1.52	1.62	1.95	6.38%	5.56%	8.73%
PPL Corp	PPL	30.88	1.65	1.66	1.80	2.74%	5.56%	10.68%
Sempra Energy	SRE	131.39	3.87	4.20	5.25	7.72%	5.56%	8.98%
Southern Co.	SO	53.47	2.46	2.54	2.78	3.06%	5.56%	10.10%
WEC Energy Group	WEC	79.95	2.36	2.50	3.00	6.27%	5.56%	8.78%
Xcel Energy Inc.	XEL	57.4	1.62	1.72	2.05	6.03%	5.56%	8.62%
Atmos Energy Corporation	ATO	102.30	2.10	2.24	2.70	6.42%	5.56%	7.78%
New Jersey Resources Corporation	NJR	49.06	1.17	1.21	1.33	3.20%	5.56%	7.86%
Northwest Natural Gas Company	NWN	67.33	1.93	1.97	2.20	3.75%	5.56%	8.36%
ONE Gas, Inc.	OGS	88.19	2.00	2.16	2.65	7.05%	5.56%	8.11%
South Jersey Industries, inc.	SJI	32.24	1.20	1.25	1.40	3.85%	5.56%	9.32%
Southwest Gas Corporation	SWX	84.92	2.18	2.30	2.60	4.17%	5.56%	8.17%
Spire, Inc.	SR	83.82	2.37	2.46	2.67	2.77%	5.56%	8.29%
Average						**4.94%**	**5.56%**	**8.83%**

Exhibit ___ (AEB-2R)

Proxy Group Analysis
Staff Finance Panel's Proxy Group
Credit Rating and Business Segment Data

Proxy Group Analysis
Staff Finance Panel's Proxy Group
Credit Rating and Business Segment Data

Company	Ticker	Credit Rating		Operating Income		Revenue	
		Moody's	S&P	Electric	Gas	Electric	Gas
Allete Inc.	ALE	Baa1	BBB+	97%	1%	98%	1%
Alliant Energy Corp	LNT	Baa1	A-	94%	7%	87%	11%
Ameren Corp.	AEE	Baa1	BBB+	88%	12%	86%	14%
American Electric Power Co. Inc.	AEP	Baa1	A-	100%	0%	100%	0%
Avista Corp	AVA	Baa2	BBB	100%	0%	100%	0%
Black Hills Corp	BKH	Baa2	BBB+	48%	52%	42%	57%
CMS Energy Corp	CMS	Baa1	BBB+	74%	26%	71%	28%
Consolidated Edison Inc.	ED	Baa1	A-	78%	19%	77%	17%
Dominion Energy Inc.	D	Baa2	BBB+	66%	34%	71%	27%
Duke Energy Corp.	DUK	Baa1	A-	93%	7%	93%	5%
Edison International	EIX	Baa3	BBB	100%	0%	100%	0%
Entergy Corporation	ETR	Baa2	BBB+	99%	1%	99%	1%
Eversource Energy	ES	Baa1	A-	91%	9%	89%	11%
First Energy Corp	FE	Baa3	BBB	100%	0%	100%	0%
Hawaiian Electric Industries	HE	WR	BBB-	100%	0%	100%	0%
IDACORP Inc.	IDA	Baa1	BBB	100%	0%	100%	0%
MGE Energy, Inc.	MGEE	NR	AA-	78%	22%	74%	26%
NextEra Energy, Inc.	NEE	Baa1	A-	100%	0%	100%	0%
Northwestern Corp	NWE	Baa2	BBB	84%	16%	79%	21%
OGE Energy Corp.	OGE	(P)Baa1	BBB+	100%	0%	100%	0%
Pinnacle West Capital Corp.	PNW	A3	A-	100%	0%	100%	0%
PNM Resources Inc.	PNM	Baa3	BBB+	100%	0%	100%	0%
Portland General Electric Co.	POR	A3	BBB+	100%	0%	100%	0%
PPL Corp	PPL	Baa2	A-	96%	4%	94%	5%
Sempra Energy	SRE	Baa1	BBB+	53%	47%	48%	52%
Southern Co.	SO	Baa2	A-	81%	19%	84%	12%
WEC Energy Group	WEC	Baa1	A-	59%	41%	60%	39%
Xcel Energy Inc.	XEL	Baa1	A-	87%	13%	85%	14%
Average				**88%**	**12%**	**87%**	**12%**
National Grid USA				**55%**	**45%**	**53%**	**47%**

Exhibit ___ (AEB-3R)

Response to IR DPS-894

Date of Request: July 17, 2019 Request No. DPS-894
Due Date: July 29, 2019 NG Request No. NG-1198

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID
THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
Case Nos. 19-G-0309 & 19-G-0310
Gas Utilities Rates

Request for Information

FROM: DPS Staff, Abdul Qadir

TO: National Grid, Ann E. Bulkley

SUBJECT: Net Operating Income Criterion

Request:

Note: In all interrogatories, all requests for workpapers or supporting calculations shall be construed as requesting any Word, Excel or other computer spreadsheet models in original electronic format with all formulae intact and unlocked.

Reference page 42 lines 12 to 15 of your testimony, which states, "To ensure that the proxy group consists of companies that are primarily regulated utilities, I eliminated companies with less than 70 percent of total operating income derived from regulated utility operations."

1. Is this criterion superior to the Commission's previously used revenue criterion, in which a utility must have regulated revenue that is 70% or greater of its total revenues? See for example the recent Case 16-G-0257, regarding National Fuel Gas Distribution Corporation?

2. If "yes," explain.

3. Are you aware of any public utility commission decision(s) nationwide that have accepted this criterion?

4. If "yes," provide the case number and the jurisdiction and any additional information necessary to identify and retrieve copies of the cited decisions.

Response:

1. Yes, the percentage of total operating income derived from regulated utility operations is a superior screening criterion to the Commission's previously used revenue criterion.

2. As described in more detail in Ms. Bulkley's Direct Testimony, at 45-47, the percentage of revenue derived from regulated operations can fluctuate significantly due to changes in purchased gas, purchased fuel, and purchased power costs. Those costs are typically recovered through tracking mechanisms and therefore do not impact a company's earnings or financial position. The percentage of total operating income derived from regulated utility operations is thus the superior screening criterion because it better reflects the company's overall financial position.

   This is further demonstrated by the fact that New Jersey Resources, which is generally regarded as a gas distribution company, would be excluded by the Commission's previously used revenue criterion, but not the proposed income criterion.

3. Yes, other public utility commissions have accepted this criterion.

4. For example, the Company's Massachusetts' gas affiliates were recently authorized an ROE derived from a proxy group selected using an income-based screening criterion, rather than a revenue-based screening criterion. Specifically, the Company proposed limiting the proxy group to only companies that receive "at least 60 percent of their total income from regulated natural gas utility operations." The Massachusetts Department of Public Utilities found that this was a valid screening criterion. (D.P.U. 17-170, at 268-271)

   The New Mexico Public Regulation Commission has also accepted an income-based screening criterion. Specifically, in Case No. 15-00261-UT, the Hearing Examiner found that the Company's proposed proxy group, which was derived from screening criteria that included an income-based criterion, was "most appropriate" (Corrected Recommended Decision, at 40-42). The Commission subsequently adopted the Hearing Examiner's Corrected Recommended Decision with regard to ROE.

Date of Request: July 17, 2019 Request No. DPS-895
Due Date: July 29, 2019 NG Request No. NG-1199

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID
THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
Case Nos. 19-G-0309 & 19-G-0310
Gas Utilities Rates

Request for Information

FROM: DPS Staff, Abdul Qadir

TO: National Grid, Ann E. Bulkley

SUBJECT: XIRR vs IRR Excel Formula for DCF Analysis

Request:

Note: In all interrogatories, all requests for workpapers or supporting calculations shall be construed as requesting any Word, Excel or other computer spreadsheet models in original electronic format with all formulae intact and unlocked.

Reference Exhibit___(AEB-1) column labelled "[13]" in which you used the extended version of the Internal Rate of Return (XIRR) as opposed to the regular IRR for your DCF analysis.

Reference also page 11, line 4 to line 15 in which you stated that "[t]hough the [Generic Finance Case] was never formally adopted by the Commission, it has served as a touchstone for the Commission's ROE determinations for more than 25 years." Then you continued, "the methodologies that I have applied to estimate the return on equity are consistent with Commission precedent since the RD [Recommended Decision] in the [Generic Finance Case]."

1. Do the Generic Finance Case RD and Commission rate case decisions since the RD was issued rely on the IRR or XIRR for DCF analysis? Provide evidence to support your response.

2. What would have been your DCF results if you had used the IRR? Provide your response in Excel Spreadsheet with formulas intact.

3. Is your use of the "XIRR" consistent with the Generic Finance Case RD and Commission's rate case decisions since the RD was issued? Provide evidence to support your response.

4. Are you aware of any public utility commission decision nationwide that have accepted the use of the XIRR formula? If your response is "yes," then provide the case number(s) and the jurisdiction(s) and other information sufficient to identify and retrieve copies of the cited decisions.

Response:

1. Ms. Bulkley is not aware of the Commission making an explicit determination as to which Excel functions should be applied to the DCF model.

2. Attachment 1 contains an Excel spreadsheet that uses the IRR function instead of the XIRR function. While the IRR function and XIRR function perform similar functions, the XIRR allows additional inputs as to the timing of cash flows. Due to the reduced functionality of the IRR function relative to the XIRR function, the logic that pertains to the timing of cash flows cannot be used with the IRR function and therefore the IRR-derived results in Attachment 1 are oversimplified and imprecise.

3. As described in response to part 1 above, Ms. Bulkley is unaware whether the Generic Finance Case RD relied on the IRR or XIRR formula. As described in Ms. Bulkley's direct testimony, page 11, lines 12-15, the models used in Ms. Bulkley's analysis "extend the principles advanced in the RD in the GFP to best practices in financial analysis and current capital market conditions, as was contemplated in the RD." As described in response to part 4 below, the XIRR function, which reflects a more nuanced set of assumptions than the IRR function, represents best practices in financial analysis.

4. Ms. Bulkley is not aware of any regulatory commission making an explicit determination as to which Excel functions should be applied to the DCF model. Many public utility commissions nationwide have accepted the practice of escalating the dividend yield within the Constant Growth Discounted Cash Flow model by one-half of the projected growth rate.[1] This adjustment is intended to reflect the fact that utility companies tend to increase their dividend payments at different times throughout the year, so a mid-year convention is accepted. The XIRR formula is intended to reflect a similar phenomenon, which assumes a mid-year convention for cash flow payments.

   The XIRR formula allows the analysis to better account for the timing of when cash flows (*i.e.*, dividend payments) are available to investors in utility stock. Utility stocks typically pay dividends on a quarterly basis throughout the year. As constructed on Exhibit___(AEB-1), the XIRR formula reflects the assumption that dividends are paid, on average, halfway through the year (*i.e.*, two dividends are paid in the first half of the year, and two dividends are paid in the second half of the year). The IRR formula, in contrast, assumes that cash flows are only paid at the end of the period. As shown in Attachment 1, that assumption within the IRR formula lowers the results of the DCF analysis by 11 to 21 basis points.

Name of Respondent: Ann E. Bulkley

Date of Reply: July 29, 2019

[1] See, for example, the decision recently received by the Company's Massachusetts' gas affiliates, D.P.U. 17-107, at 282, 285, and 292.

Exhibit ___ (AEB-4R)

Proxy Group Analysis
Staff Finance Panel's Proxy Group
Value Line Risk Measures

Proxy Group Analysis
Staff Finance Panel's Proxy Group
Value Line Risk Measures

Company	Ticker	As of June 2019 Safety	As of June 2019 Price Stability	As of March 2016 Safety	As of March 2016 Price Stability
Allete Inc.	ALE	2	95	2	95
Alliant Energy Corp	LNT	2	100	2	100
Ameren Corp.	AEE	2	95	2	95
American Electric Power Co. Inc.	AEP	1	100	2	100
Avista Corp	AVA	2	90	2	95
Black Hills Corp	BKH	2	80	2	85
CMS Energy Corp	CMS	2	100	2	100
Consolidated Edison Inc.	ED	1	100	1	100
Dominion Energy Inc.	D	2	100	2	100
Duke Energy Corp.	DUK	2	100	2	100
Edison International	EIX	3	85	2	100
Entergy Corporation	ETR	3	95	3	95
Eversource Energy	ES	1	100	1	100
First Energy Corp	FE	2	90	3	90
Hawaiian Electric Industries	HE	2	95	2	90
IDACORP Inc.	IDA	2	95	2	95
MGE Energy, Inc.	MGEE	1	85	1	95
NextEra Energy, Inc.	NEE	1	100	2	100
Northwestern Corp	NWE	2	95	3	95
OGE Energy Corp.	OGE	2	95	2	85
Pinnacle West Capital Corp.	PNW	1	100	1	100
PNM Resources Inc.	PNM	3	85	3	85
Portland General Electric Co.	POR	2	95	2	100
PPL Corp	PPL	2	95	2	100
Sempra Energy	SRE	2	95	3	100
Southern Co.	SO	2	100	2	100
WEC Energy Group	WEC	1	95	1	100
Xcel Energy Inc.	XEL	1	100	1	100
Average		**1.82**	**95**	**1.96**	**96**
Change Between Cases		**-7.27%**	**-1.48%**		